

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

7876

Nombre de lignes df clean	7876
Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5513
Taille dataset test	2362
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

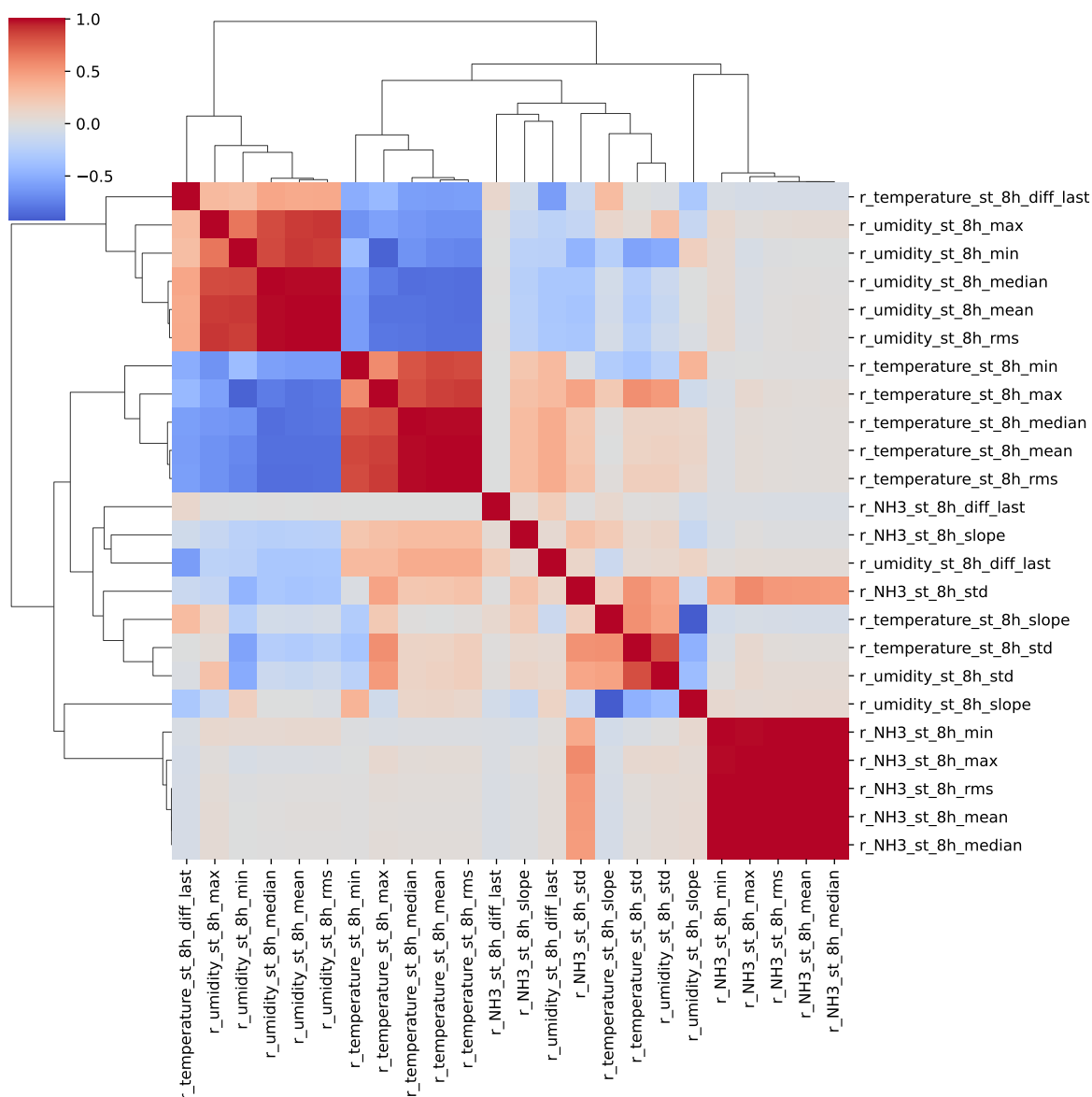
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_st_8h_max, r_NH3_st_8h_mean, r_NH3_st_8h_median, r_NH3_st_8h_rms, r_NH3_st_8h_min ; **cluster 7:** r_NH3_st_8h_std ; **cluster 8:** r_NH3_st_8h_diff_last ; **cluster 9:** r_NH3_st_8h_slope ; **cluster 10:** r_temperature_st_8h_max, r_umidity_st_8h_max, r_umidity_st_8h_median, r_temperature_st_8h_median, r_umidity_st_8h_rms, r_temperature_st_8h_mean, r_umidity_st_8h_min, r_temperature_st_8h_rms, r_temperature_st_8h_min, r_umidity_st_8h_mean ; **cluster 11:** r_temperature_st_8h_std,

r_umidity_st_8h_std ; **cluster 12:** r_temperature_st_8h_diff_last ; **cluster 13:** r_temperature_st_8h_slope, r_umidity_st_8h_slope ; **cluster 14:** r_umidity_st_8h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Chooosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_max, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_temperature_st_8h_max, r_temperature_st_8h_std, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_8h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_max, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_tempera-

ture_st_8h_max, r_temperature_st_8h_std, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_8h_diff_last

Len(Features__X): 14

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

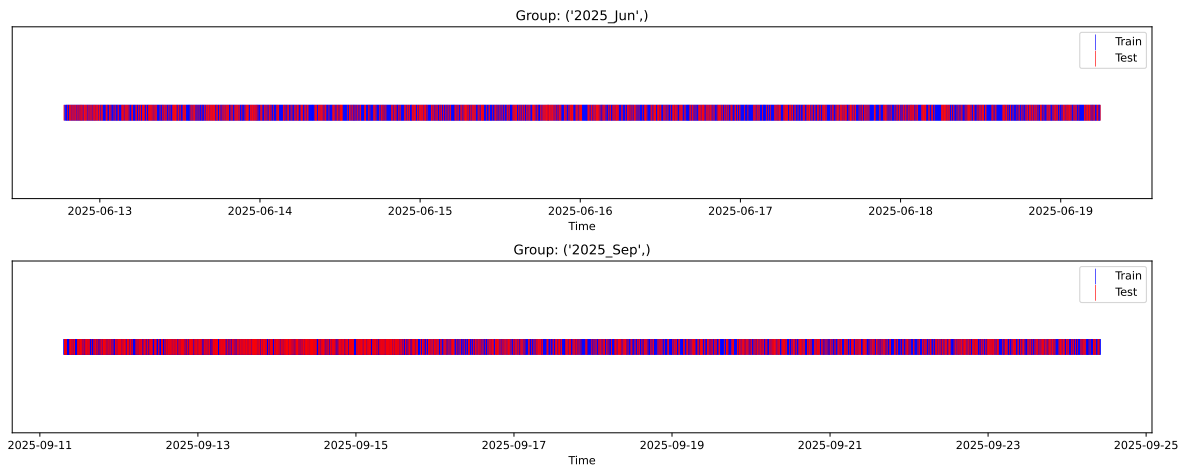
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5513 samples Test: 2363 samples



3.3.2 Internal CV Folds

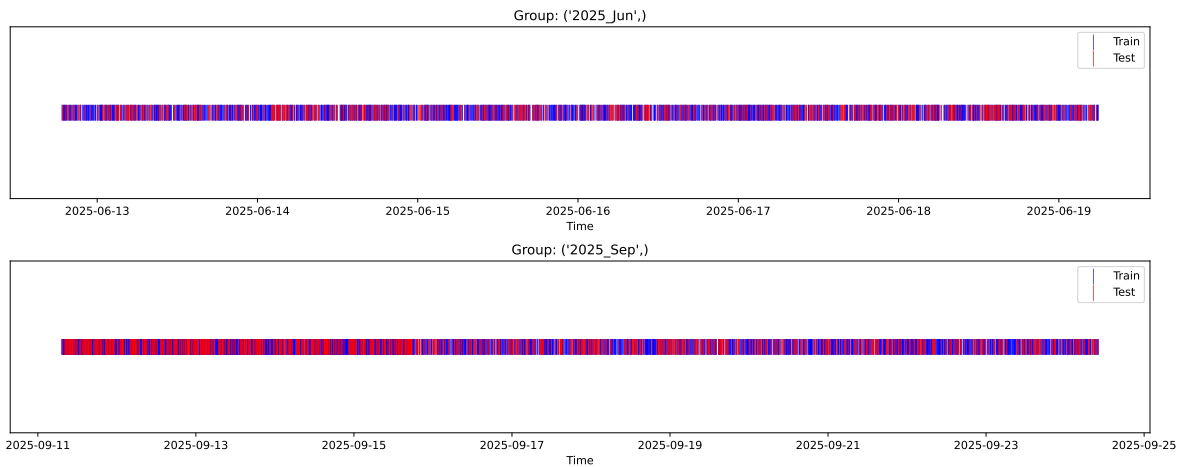
Fold 1/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 4/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.354, std=0.811, min=0.959, max=13.845
 y_{test} : mean=2.358, std=0.777, min=1.068, max=7.822

Fold 0 — y_{test} : mean=2.35, std=0.80, y_{train} : mean=2.35, std=0.82

Fold 1 — y_{test} : mean=2.32, std=0.78, y_{train} : mean=2.37, std=0.82

Fold 2 — y_{test} : mean=2.35, std=0.85, y_{train} : mean=2.35, std=0.79

Fold 3 — y_test: mean=2.36, std=0.78, y_train: mean=2.35, std=0.83

Fold 4 — y_test: mean=2.32, std=0.75, y_train: mean=2.37, std=0.84

train scores CV: [0.49740621 0.50158872 0.51478891 0.50191857 0.49530843]

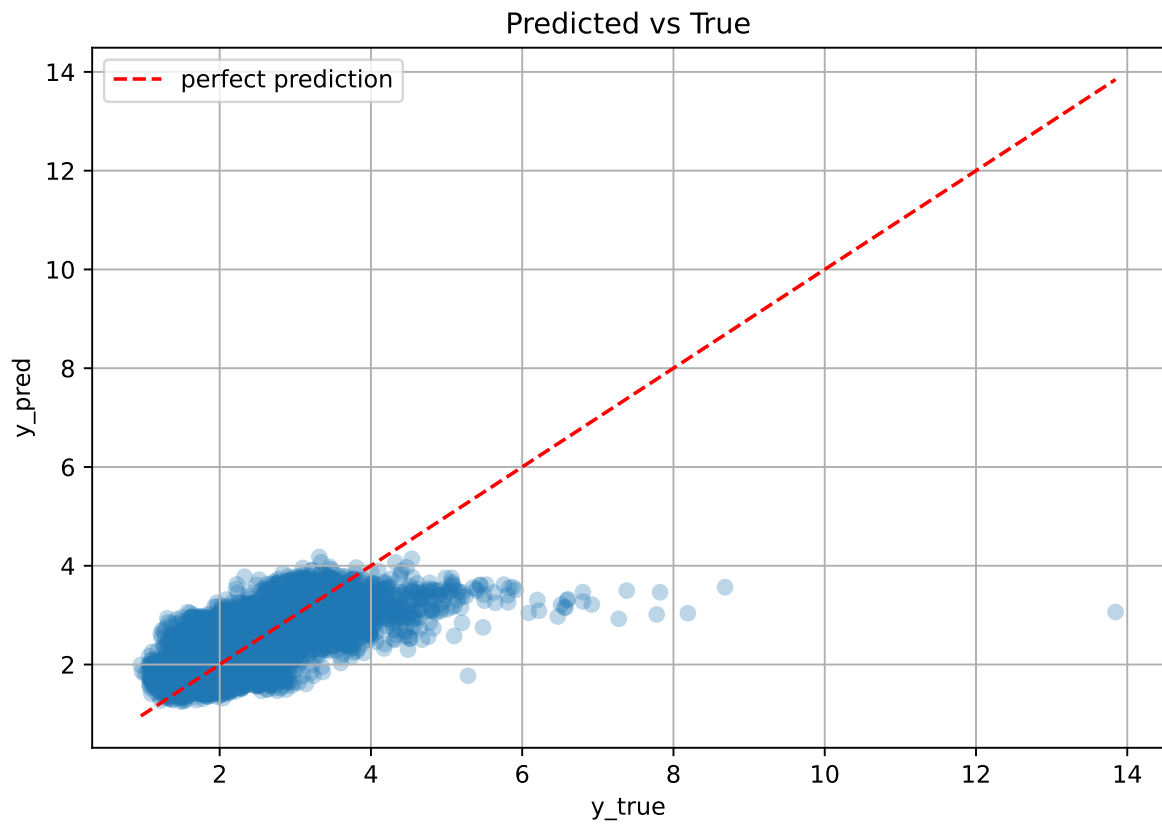
test scores CV: [0.50843353 0.4968427 0.47072614 0.49698726 0.5133202]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5022	0.4973	0.0147	0.5279	0.5291	0.0318	1.064	1.0099

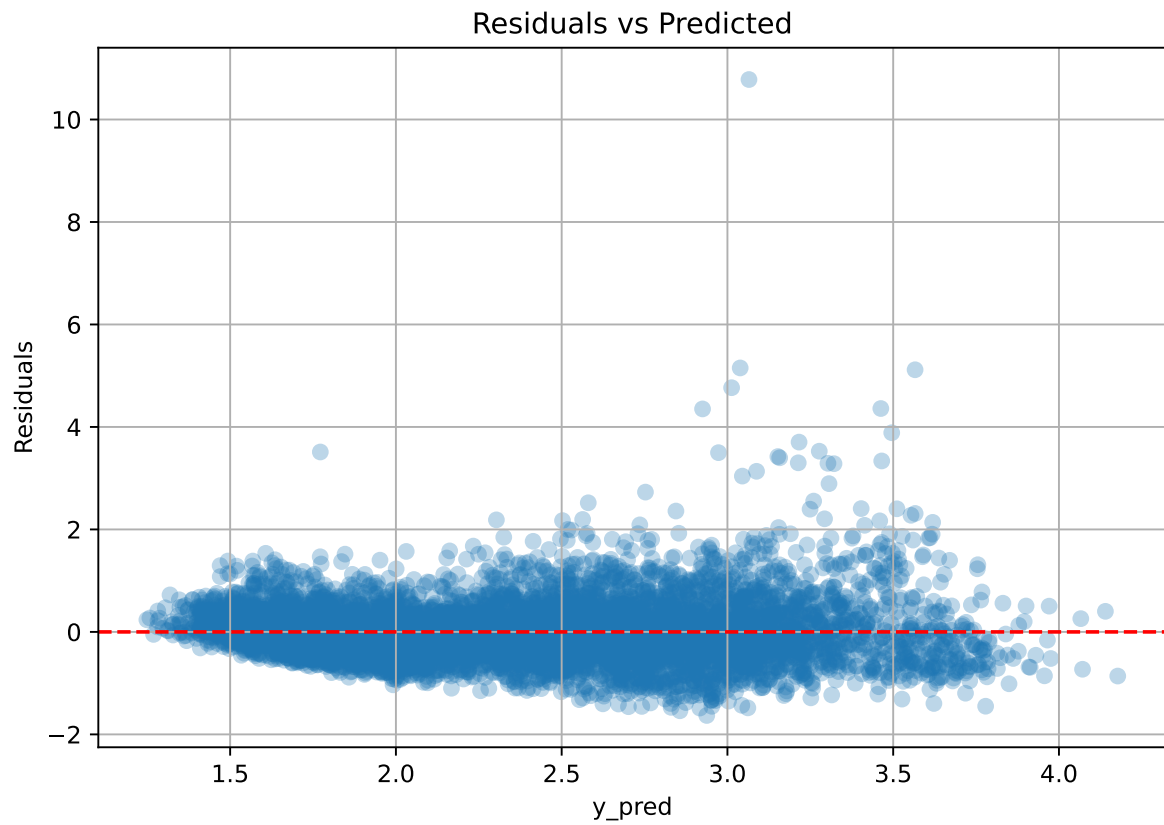
Coefficients:

coef_r_CO2 = -0.0490, coef_r_NH3 = -0.5095, coef_r_NH3_st_8h_diff_last = 0.0334, coef_r_NH3_st_8h_max = 0.4796, coef_r_NH3_st_8h_slope = 0.1266, coef_r_NH3_st_8h_std = -0.0428, coef_r_THI = 0.4556, coef_r_temperature = 0.2345, coef_r_temperature_st_8h_diff_last = 0.1161, coef_r_temperature_st_8h_max = -0.5902, coef_r_temperature_st_8h_slope = -0.5709, coef_r_temperature_st_8h_std = 0.2317, coef_r_umidity = -0.5334, coef_r_umidity_st_8h_diff_last = -0.0482, intercept = 2.3540,

3.4.2 Scatter



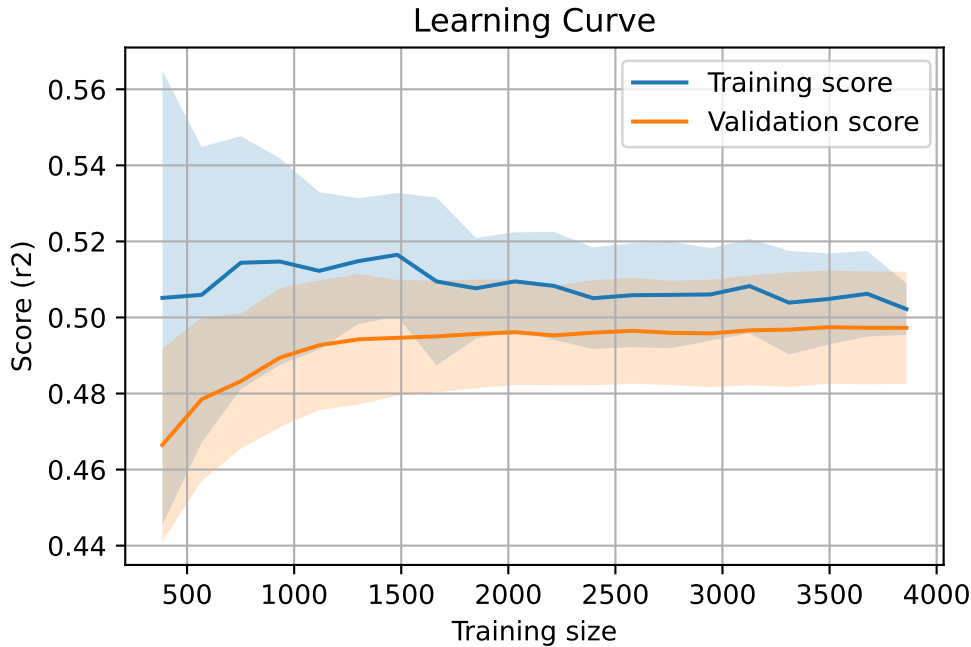
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7876, `Len(training)` : 5514,

`Len(training_real)` : 3860, `Len(test_of_training)` : 1654, `Len(test)` : 2362



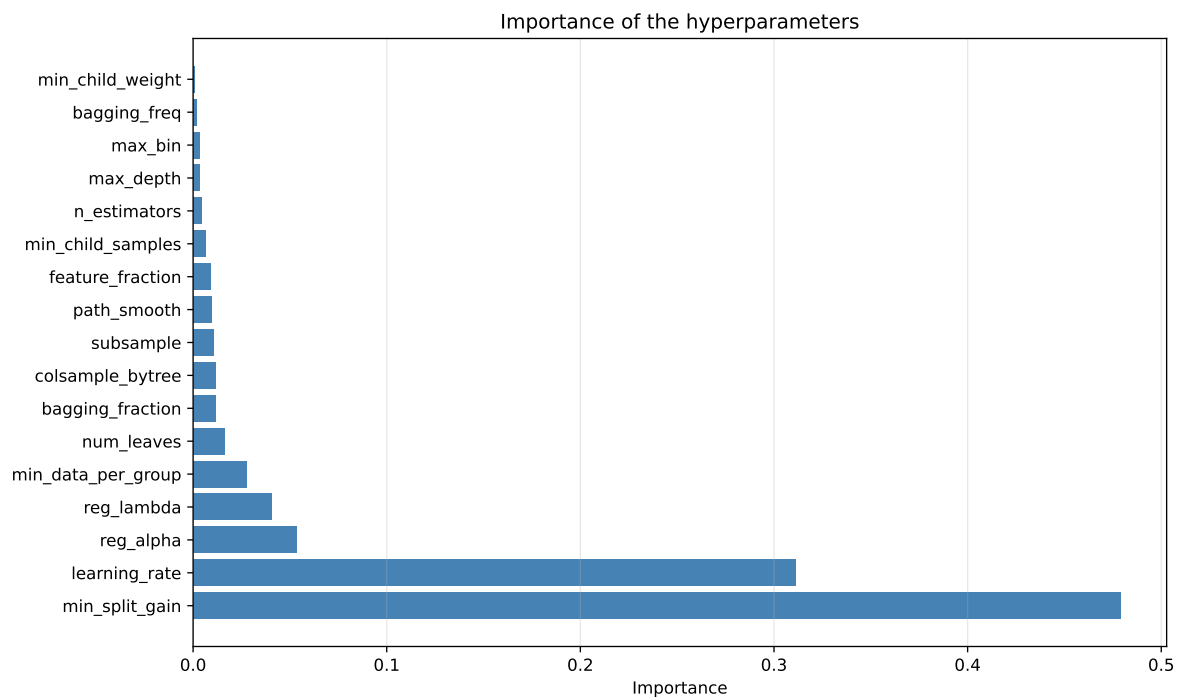
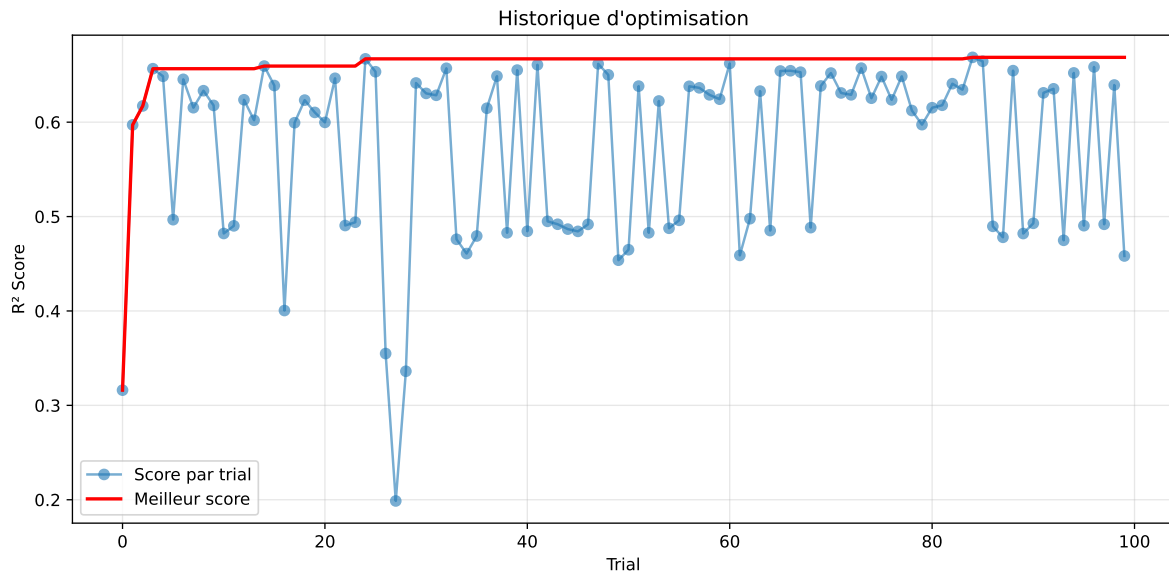
3.5 Model : LGBM

3.5.1 Gridsearch

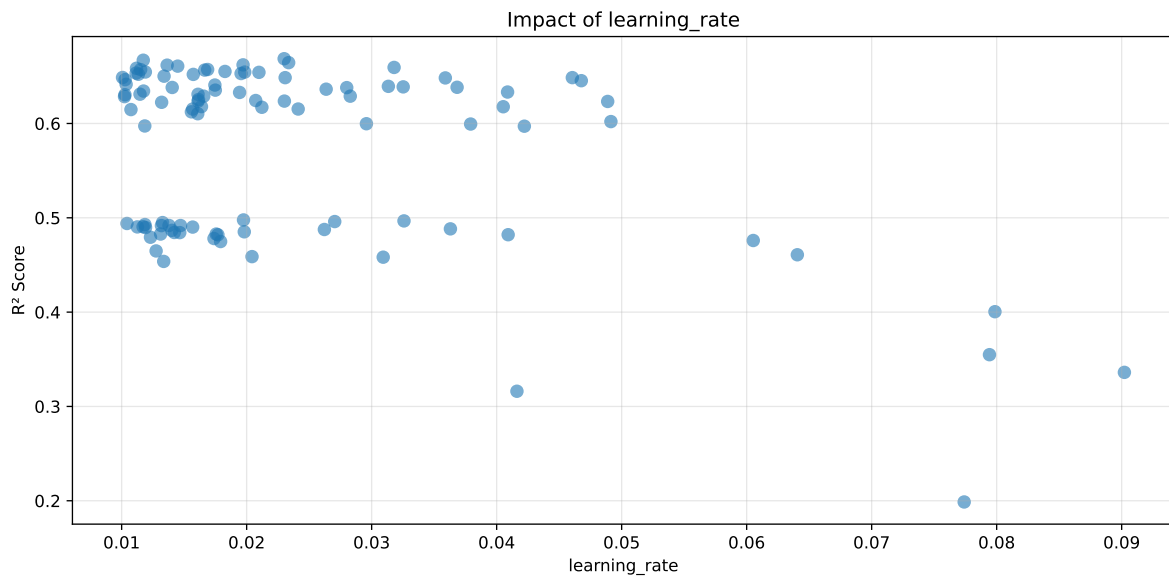
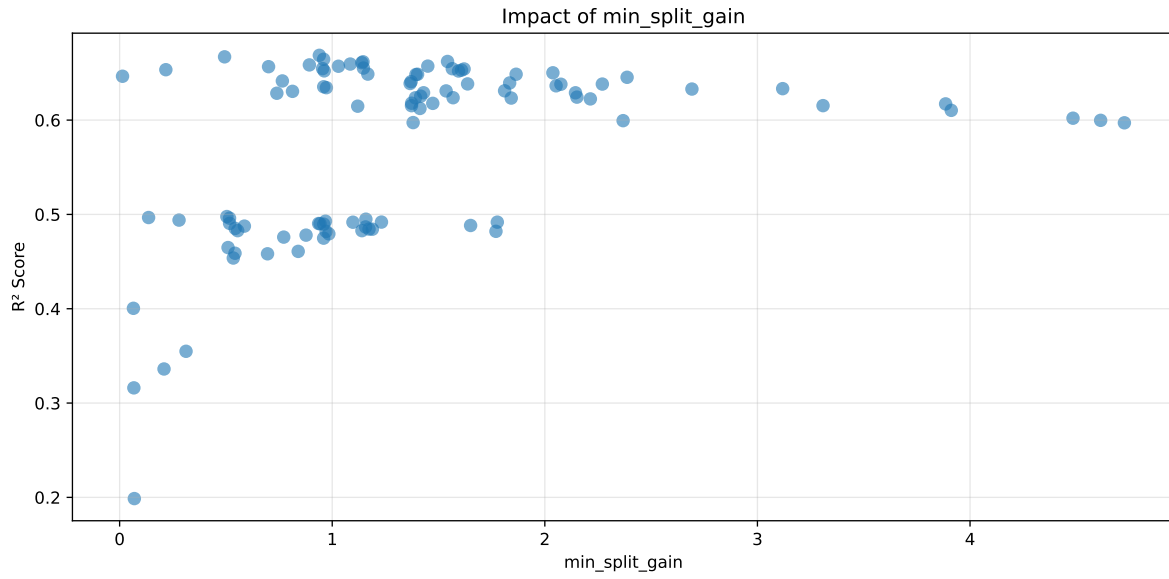
Distribution y_{train} vs y_{test} : y_{train} : mean=2.354, std=0.811, min=0.959, max=13.845
 y_{test} : mean=2.358, std=0.777, min=1.068, max=7.822

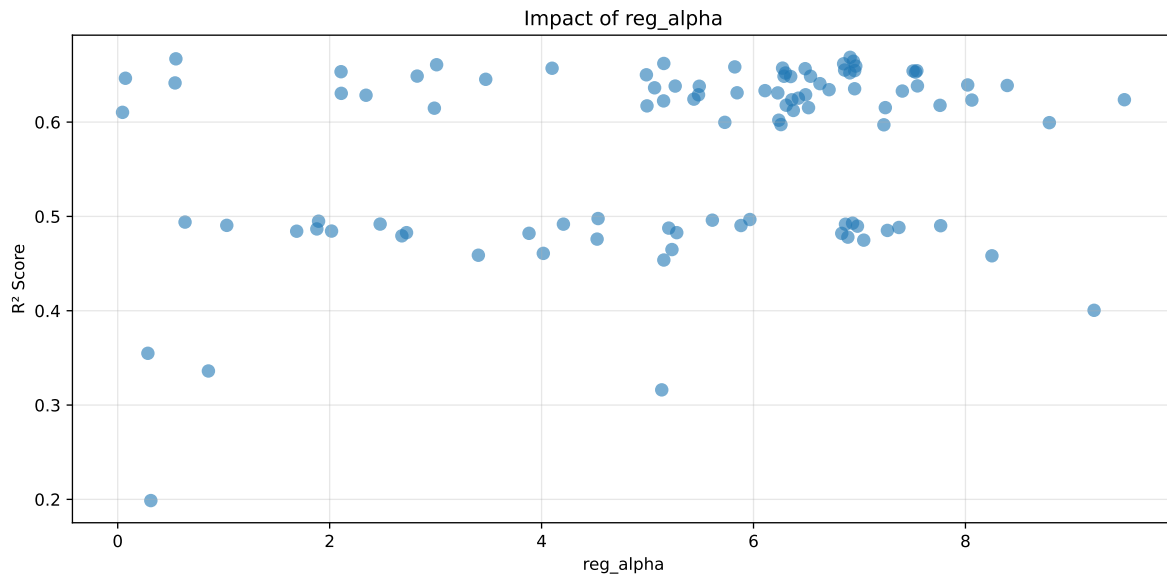
===== TOP Importance FEATURES =====

feature	importance
r_CO2	556
r_NH3_st_8h_max	493
r_umidity	334
r_temperature_st_8h_max	305
r_temperature_st_8h_slope	262
r_temperature_st_8h_diff_last	253
r_NH3_st_8h_slope	248
r_NH3	215
r_temperature	165
r_temperature_st_8h_std	154
r_NH3_st_8h_diff_last	140
r_THI	116
r_umidity_st_8h_diff_last	98
r_NH3_st_8h_std	88



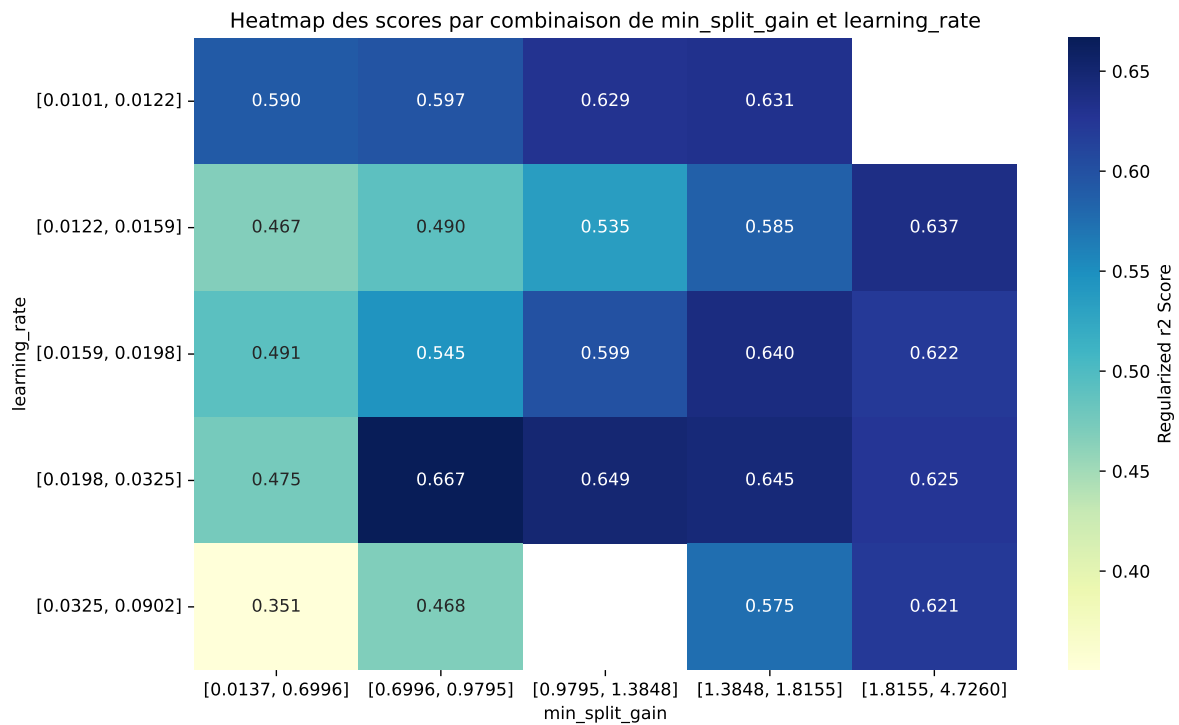
Paramètres avec >5% d'importance : ['min_split_gain', 'learning_rate', 'reg_alpha']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7019	0.6687	0.697	0.0195	0.6687	0.7182	0.0495	1.0741	1.0498
2	0.6998	0.6671	0.6925	0.024	0.6671	0.7106	0.0435	1.0652	1.0491
3	0.6949	0.6645	0.6932	0.0218	0.6645	0.7168	0.0523	1.0787	1.0458
4	0.6942	0.6622	0.6902	0.0185	0.6622	0.713	0.0508	1.0767	1.0484
5	0.6949	0.6618	0.6901	0.0201	0.6618	0.7127	0.0509	1.0768	1.05
6	0.687	0.6608	0.6886	0.0208	0.6608	0.7096	0.0488	1.0738	1.0396
7	0.6893	0.6594	0.6873	0.0186	0.6594	0.7067	0.0473	1.0717	1.0454
8	0.6891	0.6585	0.6872	0.02	0.6585	0.708	0.0495	1.0752	1.0464
9	0.6883	0.6572	0.6856	0.0175	0.6572	0.7044	0.0472	1.0718	1.0472
10	0.6895	0.6571	0.6852	0.0185	0.6571	0.701	0.0439	1.0667	1.0493
---	---	---	---	---	---	---	---	---	---
100	0.8792	0.7657	0.7797	0.0226	0.1987	0.8003	0.0345	1.0451	1.1481

Hyperparameters:

Rank 1: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.023000537373289557, 'reg__num_leaves': 32, 'reg__subsample': 0.8986178654110846, 'reg__colsample_bytree': 0.751076987569033, 'reg__min_child_samples': 21, 'reg__reg_alpha': 6.911859939617775, 'reg__reg_lambda': 24.186494993339018, 'reg__min_split_gain': 0.939422474542402, 'reg__path_smooth': 3.383580155187469, 'reg__min_child_weight': 0.0022172861227522556, 'reg__feature_fraction': 0.7797391345867489, 'reg__bagging_fraction': 0.7099988822852136, 'reg__bagging_freq': 2, 'reg__max_bin': 129, 'reg__min_data_per_group': 156}

Rank 2: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.011728986458021588, 'reg__num_leaves': 62, 'reg__subsample': 0.8011770981247046, 'reg__colsample_bytree': 0.820151293290218, 'reg__min_child_samples': 35, 'reg__reg_alpha': 0.5490444353866266, 'reg__reg_lambda': 16.25291980780679, 'reg__min_split_gain': 0.4936799920036524, 'reg__path_smooth': 4.830832536705467, 'reg__min_child_weight': 0.0019114437369875362, 'reg__feature_fraction': 0.823434433168755, 'reg__bagging_fraction': 0.7643083218708512, 'reg__bagging_freq': 5, 'reg__max_bin': 72, 'reg__min_data_per_group': 200}

Rank 3: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.02336037736997836, 'reg__num_leaves': 27, 'reg__subsample': 0.7136849873504723, 'reg__colsample_bytree': 0.7478388727154031, 'reg__min_child_samples': 48, 'reg__reg_alpha': 6.943153558198875, 'reg__reg_lambda': 8.906628148277386, 'reg__min_split_gain': 0.9595481704701182, 'reg__path_smooth': 4.567334162348385, 'reg__min_child_weight': 0.005129944487231937, 'reg__feature_fraction': 0.694426726771744, 'reg__bagging_fraction': 0.7184450235001608, 'reg__bagging_freq': 7, 'reg__max_bin': 134, 'reg__min_data_per_group': 176}

Rank 4: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.01971301380118102, 'reg__num_leaves': 37, 'reg__subsample': 0.7690689790561543, 'reg__colsample_bytree': 0.7832184100561022, 'reg__min_child_samples': 38, 'reg__reg_alpha': 5.15329322812698, 'reg__reg_lambda': 12.889816082948087, 'reg__min_split_gain': 1.5424111707646784, 'reg__path_smooth': 2.866771489073249, 'reg__min_child_weight': 0.022040455672523652, 'reg__feature_fraction': 0.7499590908061388, 'reg__bagging_fraction': 0.8523255652147249, 'reg__bagging_freq': 3, 'reg__max_bin': 154, 'reg__min_data_per_group': 146}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.013639201663770917, 'reg__num_leaves': 31, 'reg__subsample': 0.8577669539993837, 'reg__colsample_bytree': 0.7279003494990148, 'reg__min_child_samples': 40, 'reg__reg_alpha': 6.8503435221475675, 'reg__reg_lambda': 11.623434331837307, 'reg__min_split_gain': 1.14492719716258, 'reg__path_smooth': 3.8489807697608174, 'reg__min_child_weight': 0.004387938754683381, 'reg__feature_fraction': 0.8522180679373931, 'reg__bagging_fraction': 0.8278049382185412, 'reg__bagging_freq': 2, 'reg__max_bin': 121, 'reg__min_data_per_group': 152}

Rank 6: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.014485884316092304, 'reg__num_leaves': 58, 'reg__subsample': 0.8463550492272122, 'reg__colsample_bytree': 0.7272191181736739, 'reg__min_child_samples': 42, 'reg__reg_alpha': 3.0093944853693357, 'reg__reg_lambda': 20.8003725823142, 'reg__min_split_gain': 1.1391065551280875, 'reg__path_smooth': 3.5478806815659425, 'reg__min_child_weight': 0.003163304123442146, 'reg__feature_fraction': 0.8874153313650218, 'reg__bagging_fraction': 0.6199710447901717, 'reg__bagging_freq': 5, 'reg__max_bin': 94, 'reg__min_data_per_group': 152}

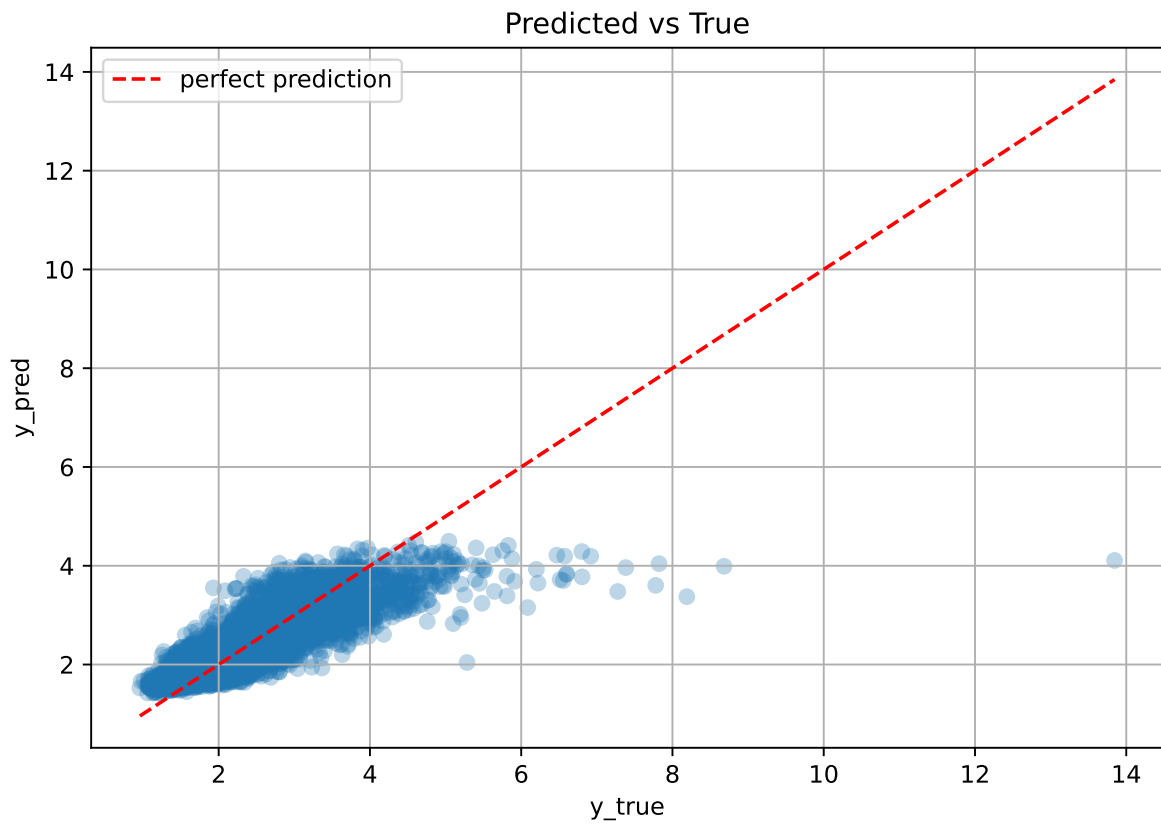
Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.03179527128320754, 'reg__num_leaves': 34, 'reg__subsample': 0.798032540282483, 'reg__colsample_bytree': 0.7547278754699147, 'reg__min_child_samples': 26, 'reg__reg_alpha': 6.964925759226865, 'reg__reg_lambda': 19.73232160324657, 'reg__min_split_gain': 1.0848313325039367, 'reg__path_smooth': 2.7431226982511108, 'reg__min_child_weight': 0.017957362083254096, 'reg__feature_fraction': 0.7891172531012258, 'reg__bagging_fraction': 0.7137368293709426, 'reg__bagging_freq': 3, 'reg__max_bin': 85, 'reg__min_data_per_group': 191}

Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.011185916544919534, 'reg__num_leaves': 31, 'reg__subsample': 0.6383400211221374, 'reg__colsample_bytree': 0.6997334411155132, 'reg__min_child_samples': 34, 'reg__reg_alpha': 5.823186082306978, 'reg__reg_lambda': 9.733161454610116, 'reg__min_split_gain': 0.8925017331980659, 'reg__path_smooth': 4.893321823840065, 'reg__min_child_weight': 0.13125442455132635, 'reg__feature_fraction': 0.7140471810127971, 'reg__bagging_fraction': 0.7501746849327732, 'reg__bagging_freq': 7, 'reg__max_bin': 127, 'reg__min_data_per_group': 164}

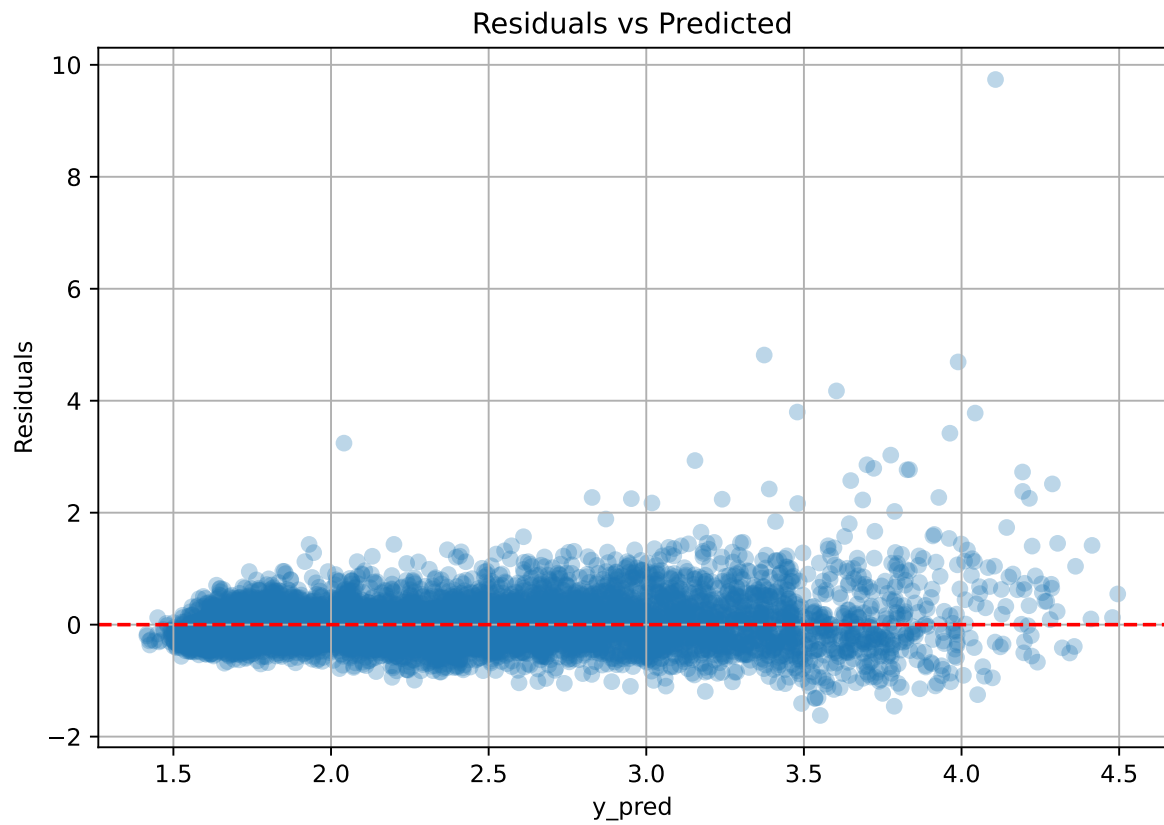
Rank 9: {'reg__n_estimators': 500, 'reg__max_depth': 4, 'reg__learning_rate': 0.01687923794664886, 'reg__num_leaves': 39, 'reg__subsample': 0.7997547890770452, 'reg__colsample_bytree': 0.8702818969352919, 'reg__min_child_samples': 31, 'reg__reg_alpha': 6.2742833519543115, 'reg__reg_lambda': 14.383915546628426, 'reg__min_split_gain': 1.449561010243608, 'reg__path_smooth': 3.85892080328348, 'reg__min_child_weight': 0.0918717693065049, 'reg__feature_fraction': 0.7511122843057267, 'reg__bagging_fraction': 0.8443683967813663, 'reg__bagging_freq': 1, 'reg__max_bin': 130, 'reg__min_data_per_group': 134}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.011518618901999865, 'reg__num_leaves': 29, 'reg__subsample': 0.8128473012641687, 'reg__colsample_bytree': 0.8312463182335733, 'reg__min_child_samples': 35, 'reg__reg_alpha': 4.09910341137915, 'reg__reg_lambda': 8.300668117792886, 'reg__min_split_gain': 1.0291877140979668, 'reg__path_smooth': 2.070205110297422, 'reg__min_child_weight': 0.003347985932004585, 'reg__feature_fraction': 0.7754302713575788, 'reg__bagging_fraction': 0.7057586668942833, 'reg__bagging_freq': 2, 'reg__max_bin': 109, 'reg__min_data_per_group': 160}

3.5.2 Scatter



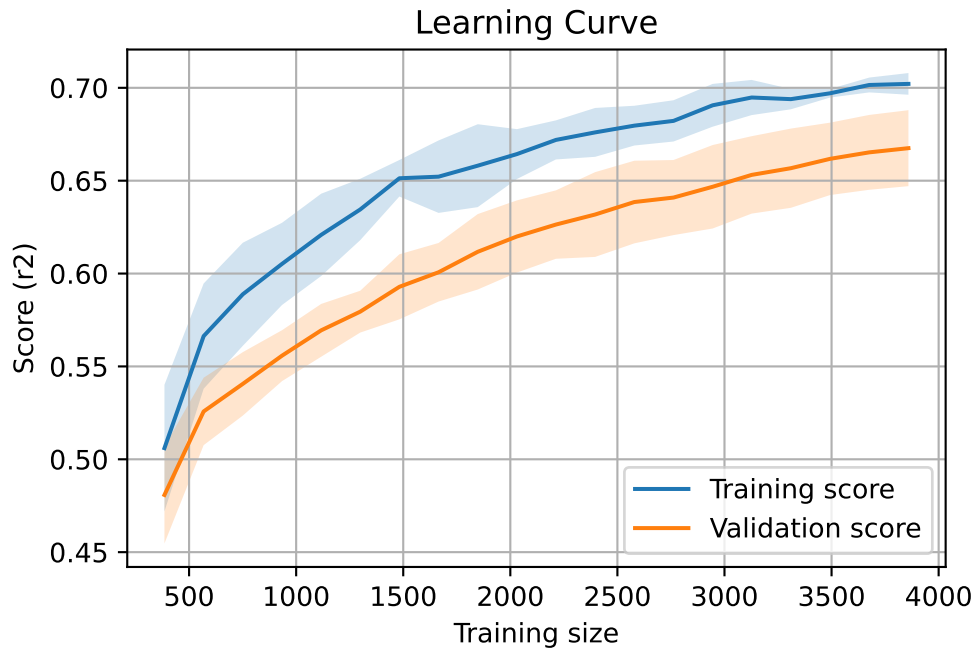
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7876, `Len(training)` : 5514,

`Len(training_real)` : 3860, `Len(test_of_training)` : 1654, `Len(test)` : 2362

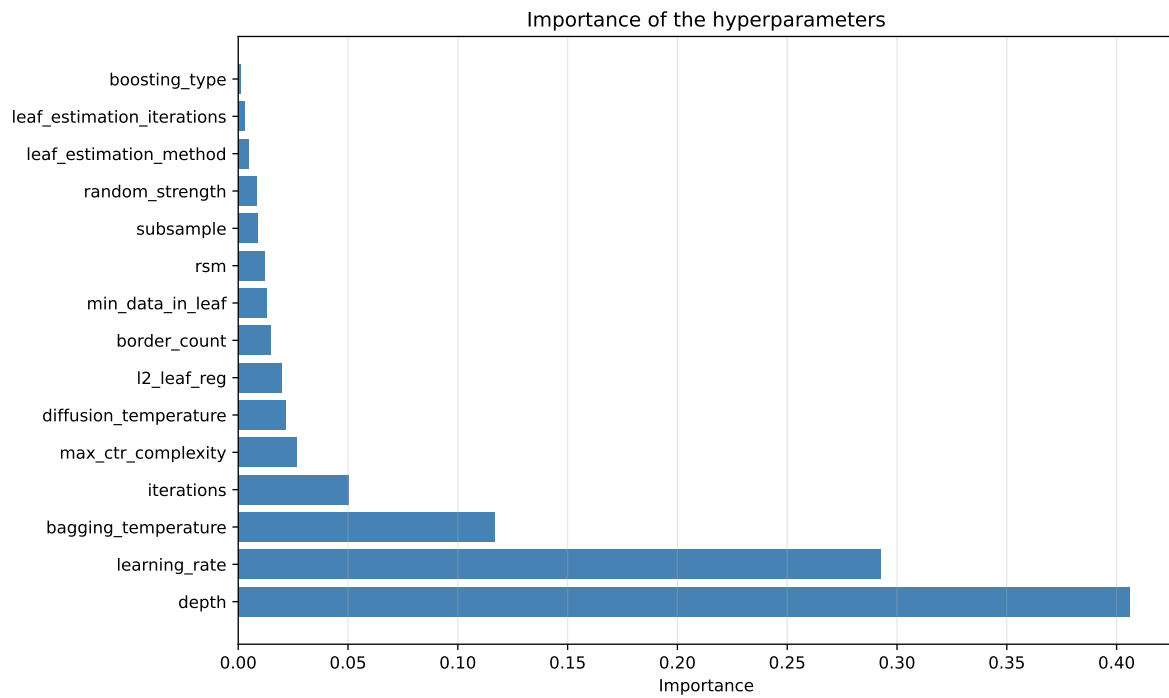
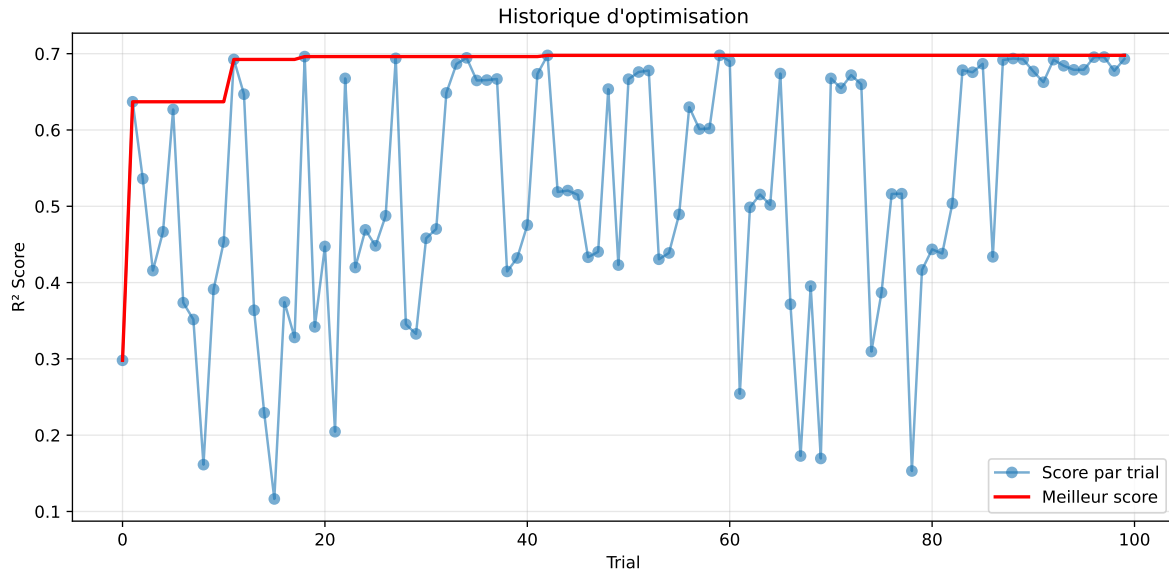


3.6 Model : CatBoost

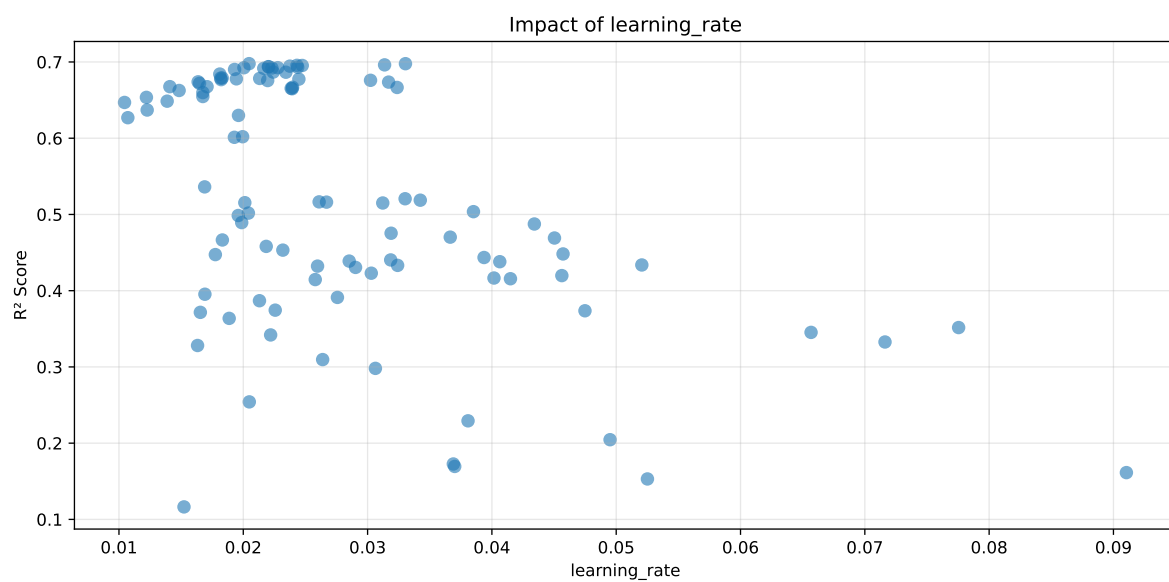
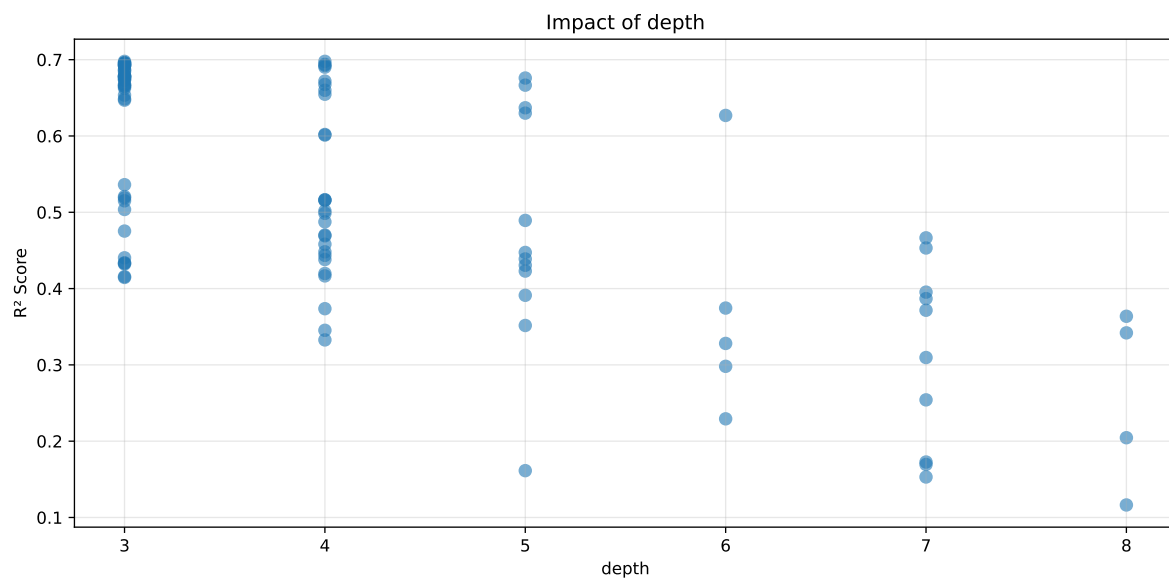
3.6.1 Gridsearch

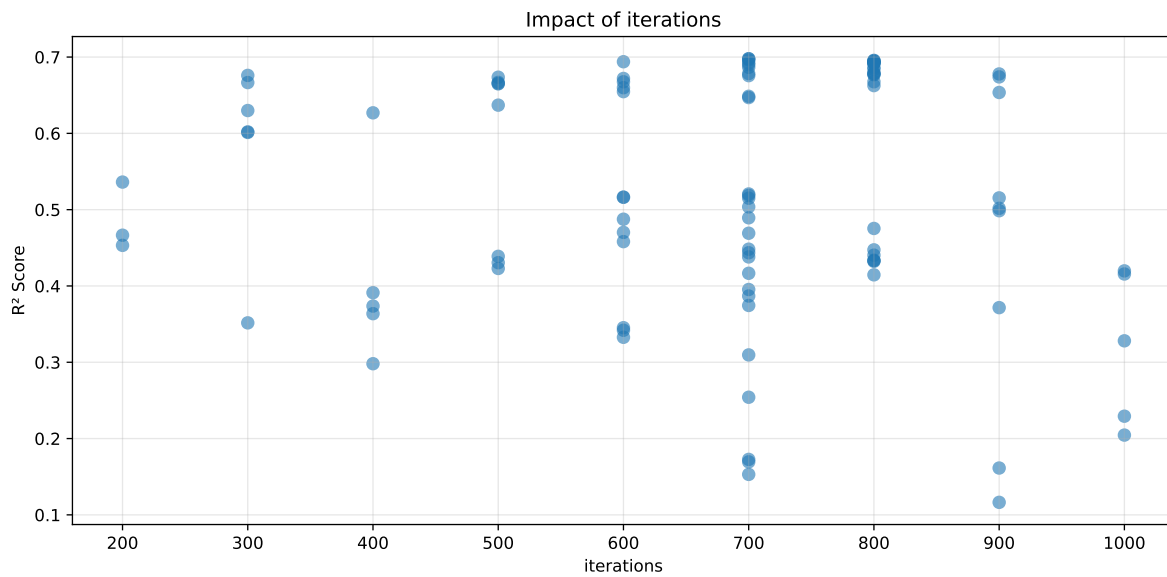
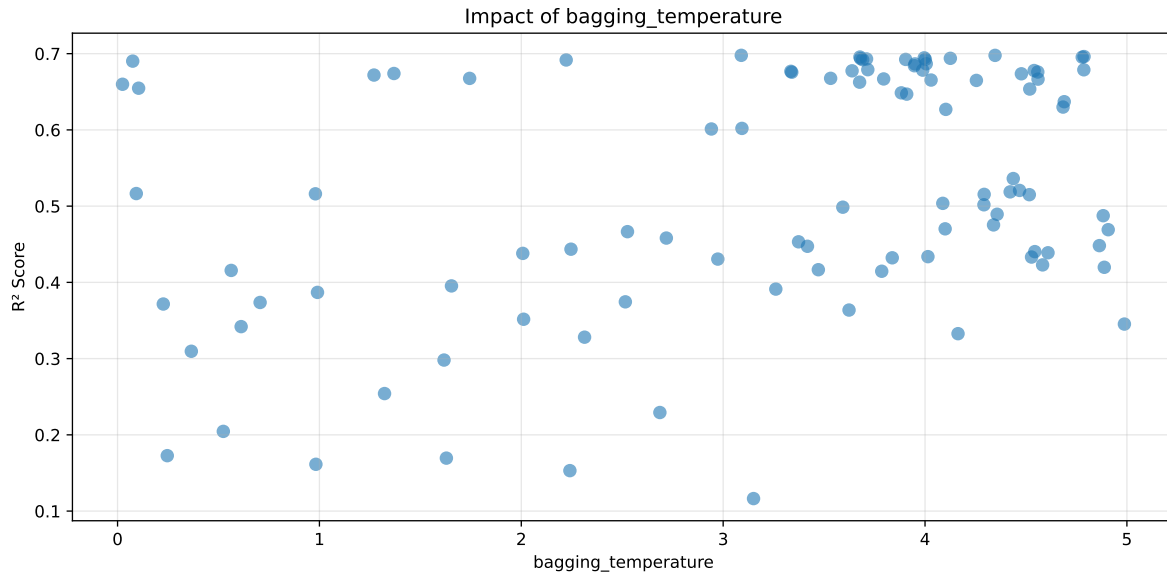
Distribution y_train vs y_test: y_train: mean=2.354, std=0.811, min=0.959, max=13.845
y_test: mean=2.358, std=0.777, min=1.068, max=7.822

===== TOP Importance FEATURES =====
feature importance r_temperature_st_8h_diff_last 19.544029 r_umidity 19.167512 r_tem-
perature_st_8h_slope 9.525463 r_CO2 9.173635 r_temperature_st_8h_max 7.700948
r_NH3_st_8h_max 7.362579 r_NH3 6.455473 r_temperature 5.052699 r_NH3_st_8h_slope
4.095075 r_umidity_st_8h_diff_last 3.287364 r_THI 2.701829 r_NH3_st_8h_std 2.671249
r_temperature_st_8h_std 2.277593 r_NH3_st_8h_diff_last 0.984550



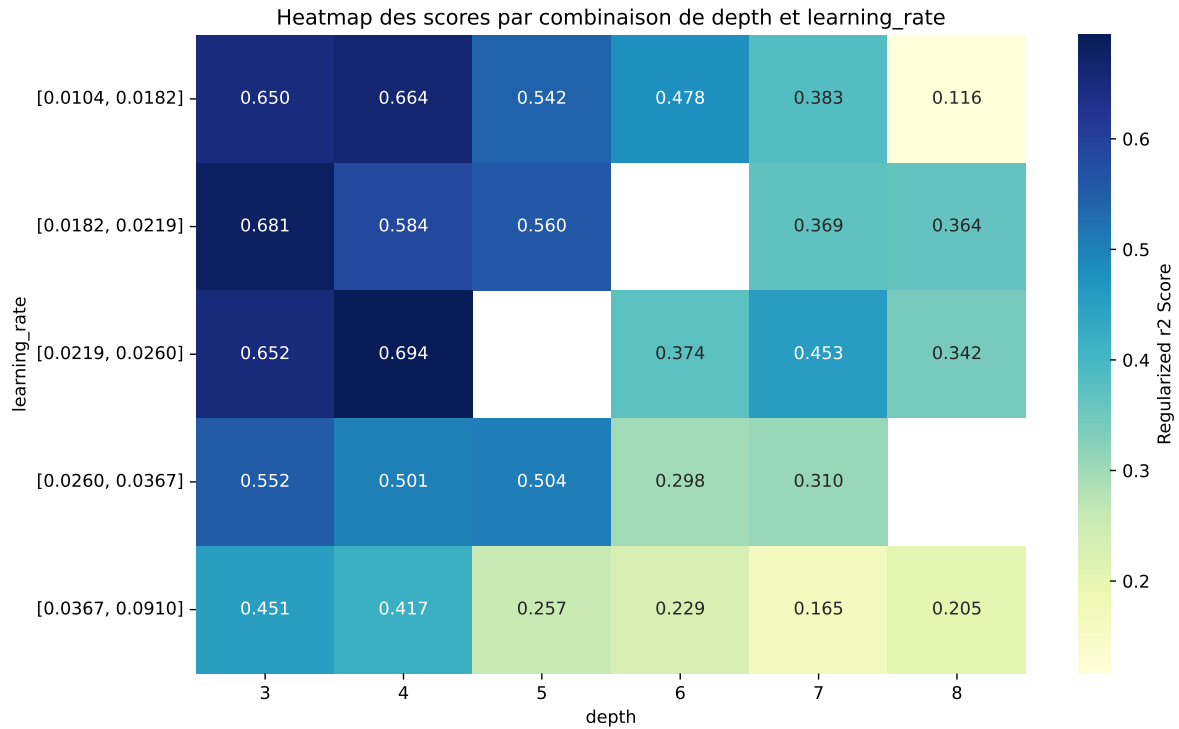
Paramètres avec >5% d'importance : ['depth', 'learning_rate', 'bagging_temperature', 'iterations']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7296	0.6978	0.7223	0.0197	0.6978	0.7274	0.0296	1.0425	1.0456
2	0.731	0.6978	0.7205	0.0194	0.6978	0.7295	0.0317	1.0455	1.0476
3	0.7272	0.6962	0.7161	0.0178	0.6962	0.7218	0.0256	1.0368	1.0445
4	0.7301	0.6956	0.7175	0.0176	0.6956	0.726	0.0304	1.0437	1.0496
5	0.7284	0.6954	0.7189	0.0196	0.6954	0.7233	0.0279	1.0402	1.0474
6	0.7262	0.6945	0.7175	0.0185	0.6945	0.7251	0.0305	1.044	1.0456
7	0.7249	0.6939	0.7181	0.018	0.6939	0.7247	0.0309	1.0445	1.0446
8	0.725	0.6936	0.7184	0.0197	0.6936	0.7219	0.0283	1.0407	1.0453
9	0.726	0.6931	0.7173	0.0184	0.6931	0.721	0.028	1.0404	1.0475
10	0.7227	0.6925	0.7132	0.0169	0.6925	0.7156	0.0231	1.0333	1.0436
---	---	---	---	---	---	---	---	---	---
100	0.8936	0.7641	0.7765	0.0227	0.1164	0.7902	0.0261	1.0341	1.1695

Hyperparameters:

Rank 1: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.02046583642296789, 'reg__l2_leaf_reg': 46.193194463651366, 'reg__bagging_temperature': 3.0897232919414606, 'reg__random_strength': 3.3354793867844554, 'reg__subsample': 0.5909596658040345, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5964499934868784, 'reg__border_count': 118, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4883.018934891075}

Rank 2: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.03304677910057024, 'reg__l2_leaf_reg': 19.98676400188481, 'reg__bagging_temperature': 4.34696685792639, 'reg__random_strength': 4.036685096386719, 'reg__subsample': 0.5438144493989591, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.5686730767017169, 'reg__border_count': 126, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 78.91547148710129}

Rank 3: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.031367500866006384, 'reg__l2_leaf_reg': 47.39334081258952, 'reg__bagging_temperature': 4.787757106866123, 'reg__random_strength': 3.6784411170783167, 'reg__subsample': 0.7906174166742982, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.7768756818446935, 'reg__border_count': 108, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 8021.3973294139305}

Rank 4: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.024318199836993445, 'reg__l2_leaf_reg': 36.56489545246352, 'reg__bagging_temperature': 3.677418067787776, 'reg__random_strength': 1.0646378827370016, 'reg__subsample': 0.8192499132603828, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.8150691811185763, 'reg__border_count': 127, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 387.9643936301168}

Rank 5: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.024743927632748974, 'reg__l2_leaf_reg': 35.87320952294603, 'reg__bagging_temperature': 4.778727555525868, 'reg__random_strength': 0.9172473650220938, 'reg__subsample': 0.8565446672461848, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.5094186398725744, 'reg__border_count': 126, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 649.8341728108201}

Rank 6: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.02374427960594468, 'reg__l2_leaf_reg': 23.88268735828883, 'reg__bagging_temperature': 3.9961256358326938, 'reg__random_strength': 1.7925318071556533, 'reg__subsample': 0.6109738251631718, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5689764732660166, 'reg__border_count': 103, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 3879.0179263791993}

Rank 7: {'reg__iterations': 600, 'reg__depth': 4, 'reg__learning_rate': 0.022017772318538426, 'reg__l2_leaf_reg': 30.347189613694418, 'reg__bagging_temperature': 4.125449674297735, 'reg__random_strength': 2.361299731618719, 'reg__subsample': 0.6488721928534725,

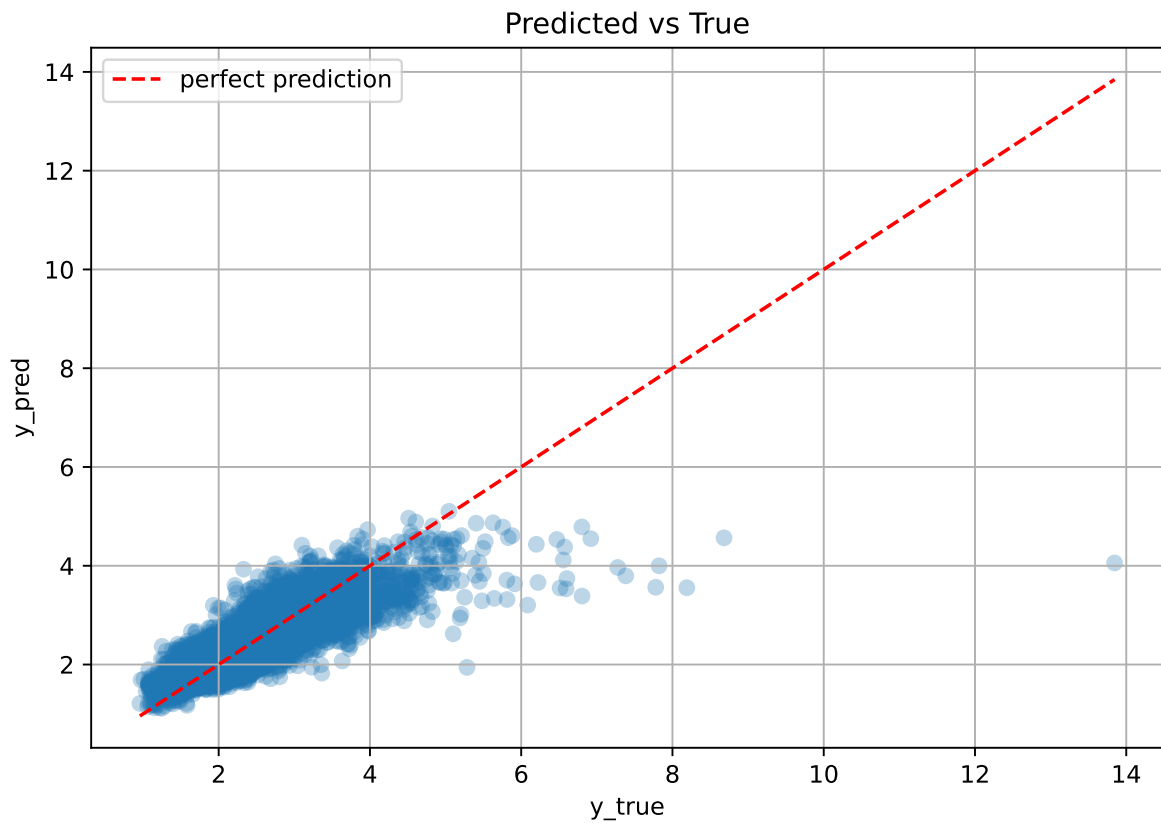
'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.8073569945289839, 'reg__border_count': 82, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 3429.0576739179846}

Rank 8: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.02204003653178722, 'reg__l2_leaf_reg': 28.736462613524566, 'reg__bagging_temperature': 3.680934872768754, 'reg__random_strength': 1.28701653902995, 'reg__subsample': 0.6532408229768796, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.638763766605899, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 3426.9666893420585}

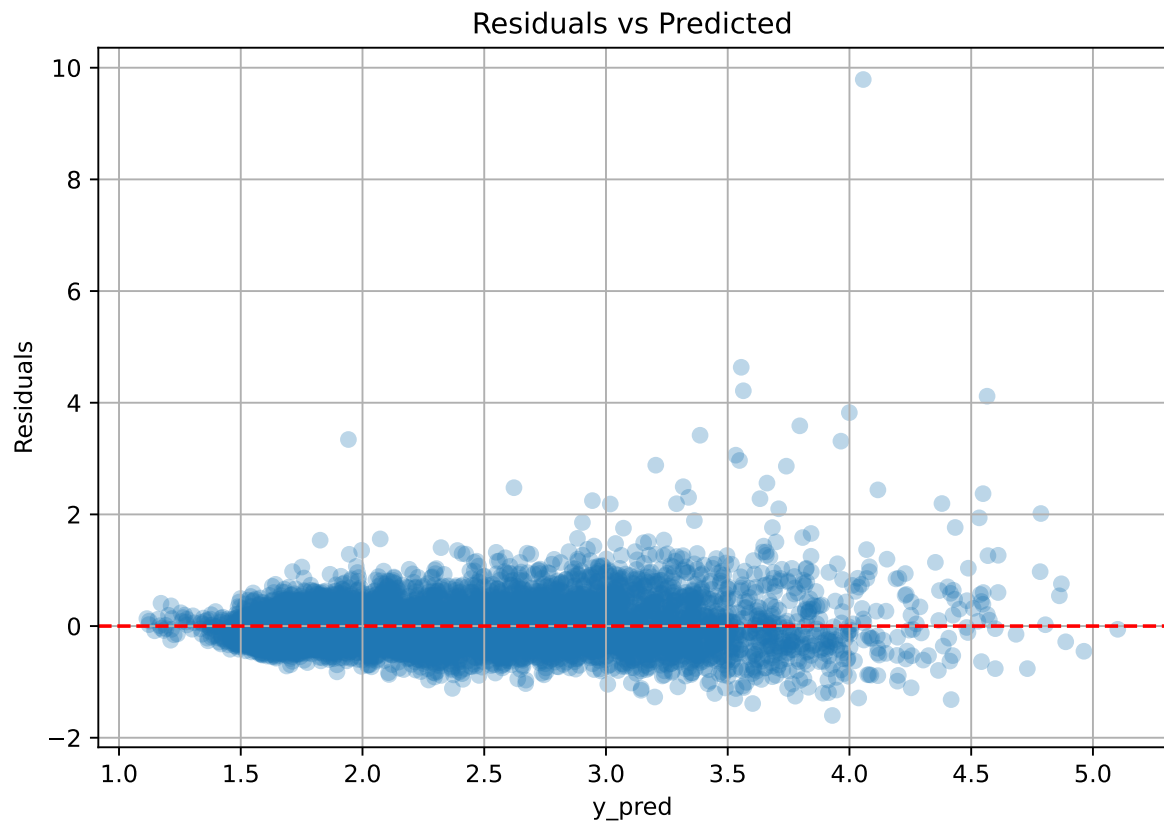
Rank 9: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.02437663884984745, 'reg__l2_leaf_reg': 37.093838296863275, 'reg__bagging_temperature': 3.7102892650671846, 'reg__random_strength': 1.0268405106550116, 'reg__subsample': 0.8692738396929105, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.5095385827319638, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 3437.8901122519887}

Rank 10: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.02277881039572601, 'reg__l2_leaf_reg': 28.211094222379145, 'reg__bagging_temperature': 4.001670871147198, 'reg__random_strength': 2.87130793348826, 'reg__subsample': 0.8003544298150977, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8455323007663151, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 8904.005763561881}

3.6.2 Scatter



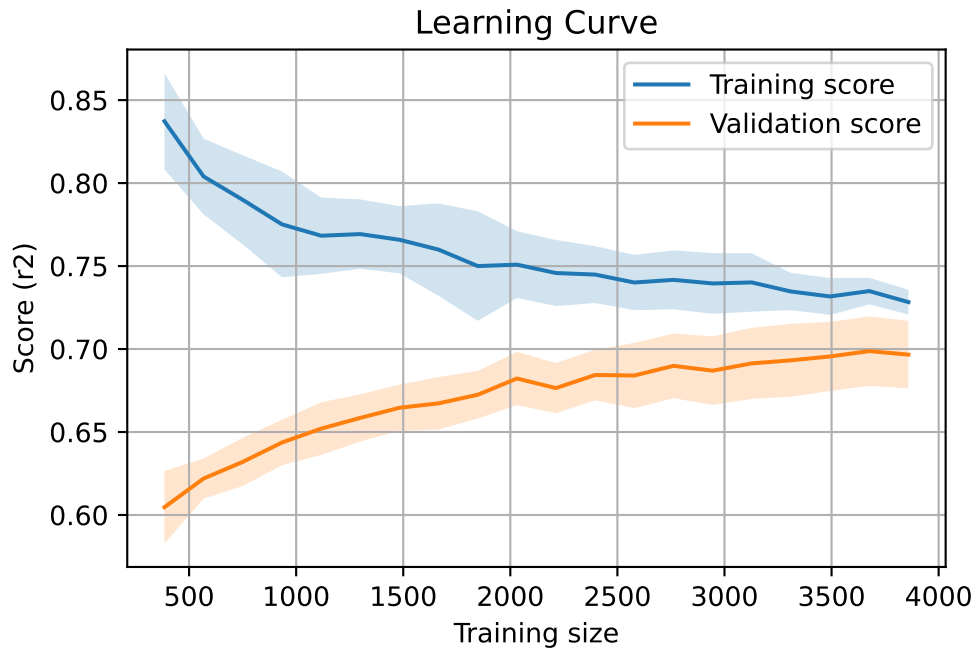
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7876, `Len(training)` : 5514,

`Len(training_real)` : 3860, `Len(test_of_training)` : 1654, `Len(test)` : 2362



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.502	0.497	0.015	0.528	0.529	0.032	1.064	1.01	nan
LGBM	0.702	0.669	0.02	0.697	0.718	0.05	1.074	1.05	0.6687
CatBoost	0.73	0.698	0.02	0.722	0.727	0.03	1.042	1.046	0.6978

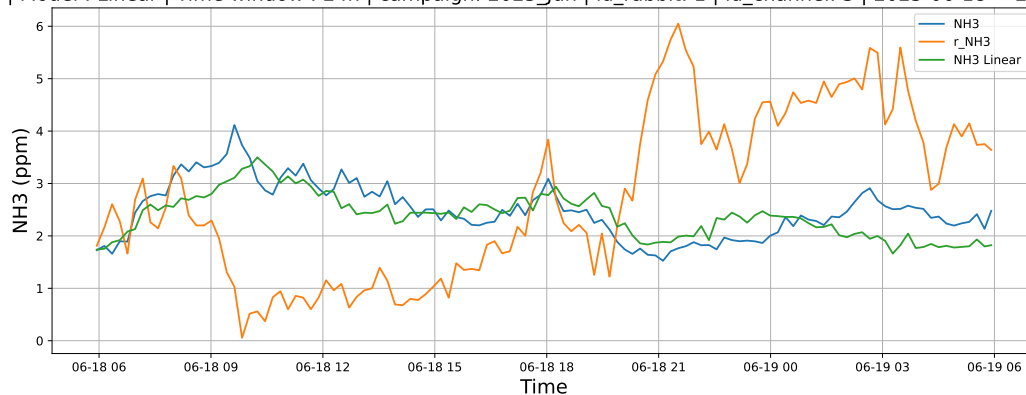
4 Plot

4.1 Standalone plots

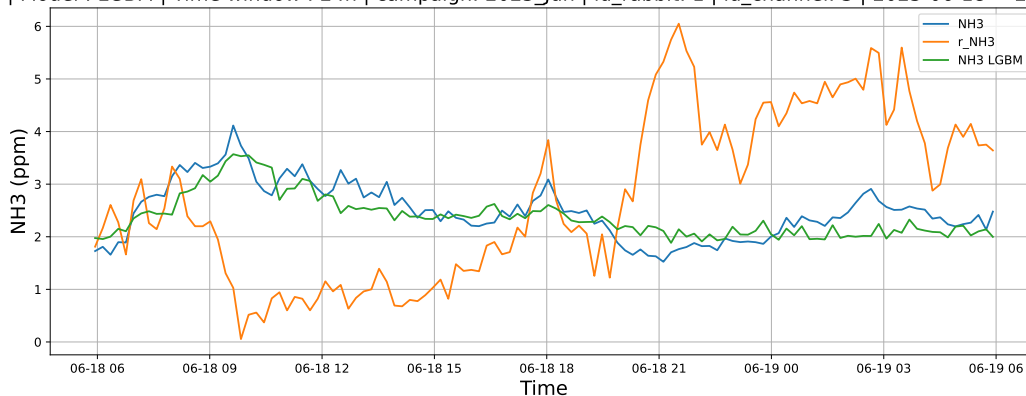
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

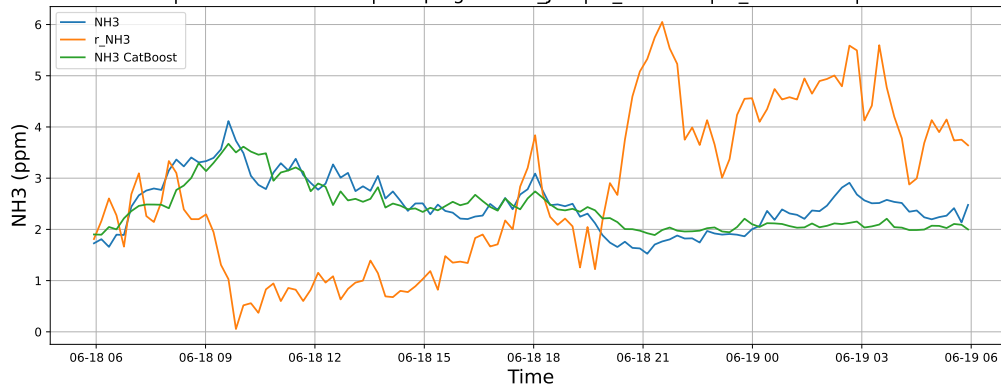
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



4.1.1.2 Time window : 48

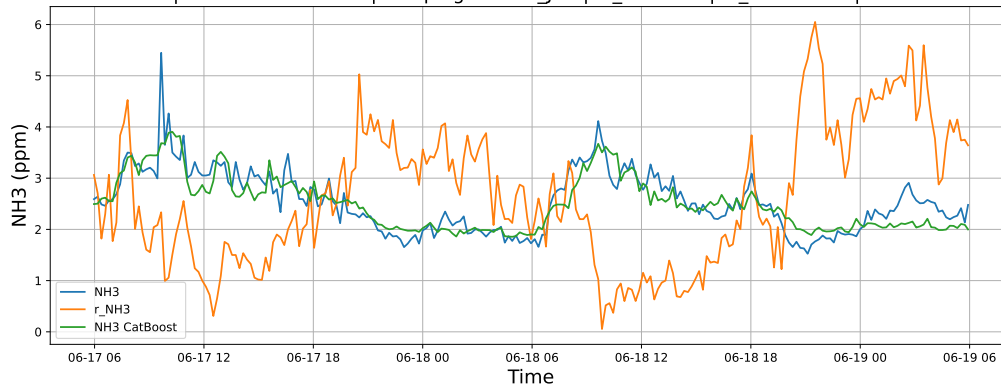
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

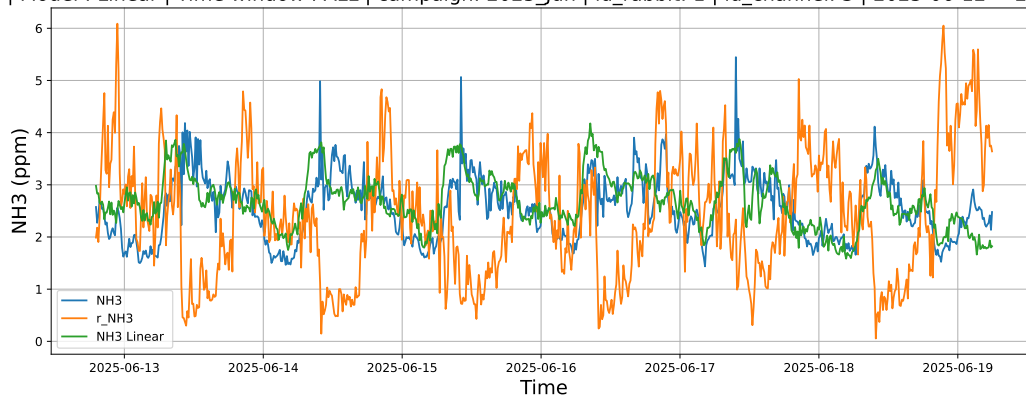


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

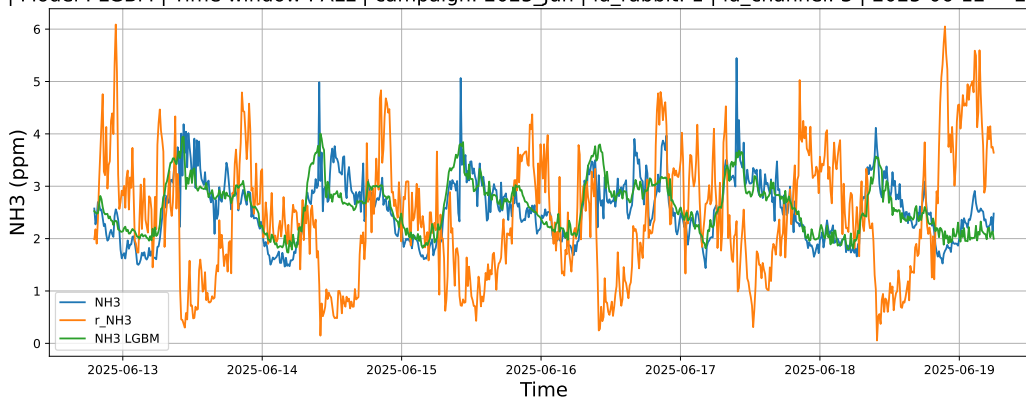


4.1.1.3 Time window : ALL

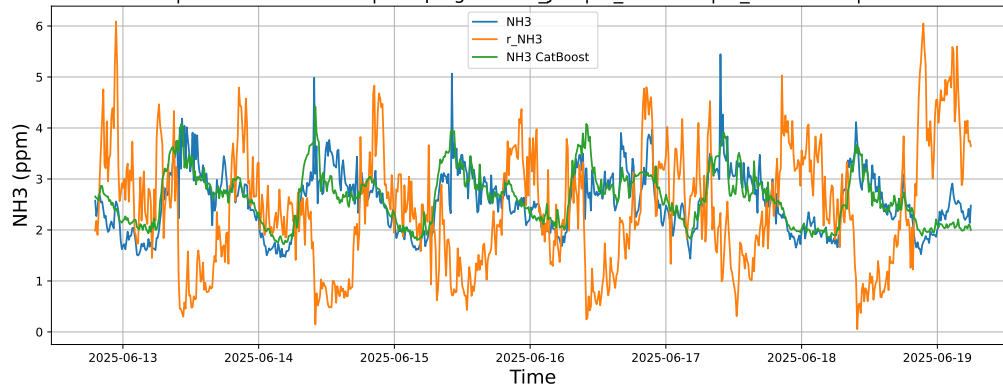
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



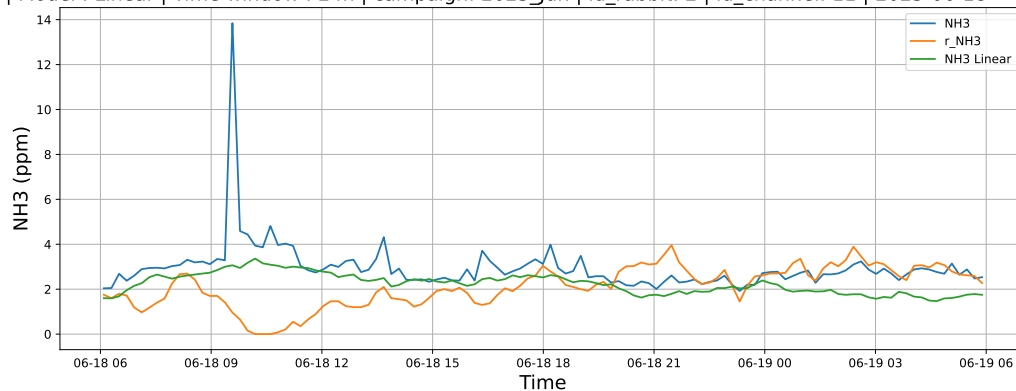
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



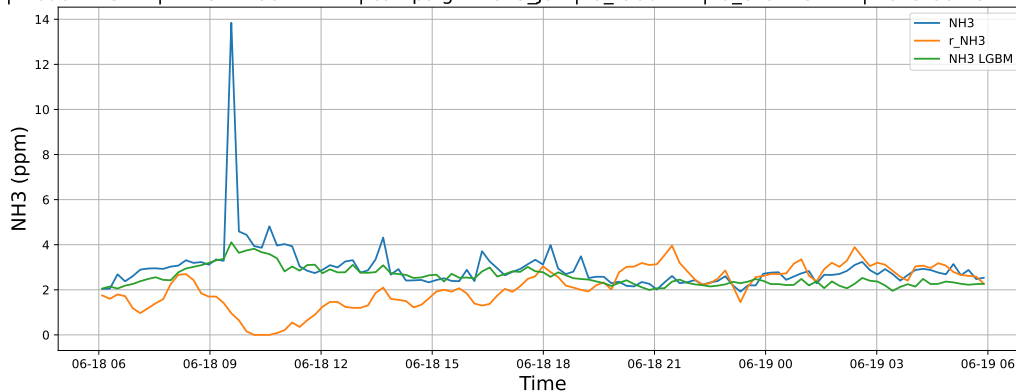
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

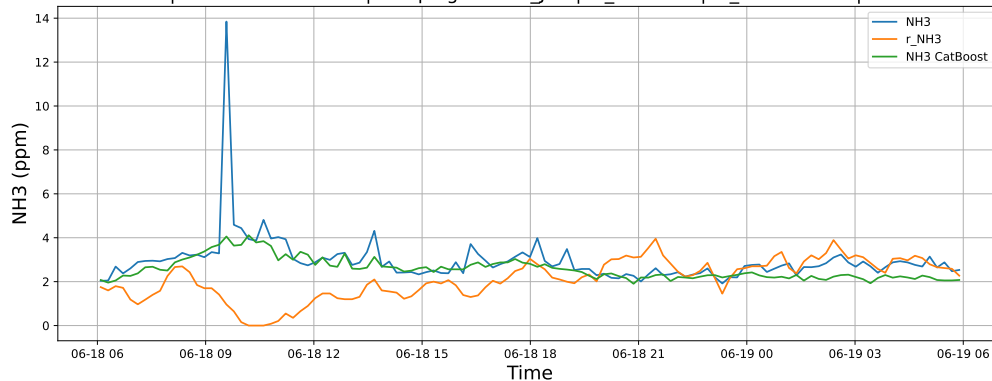
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

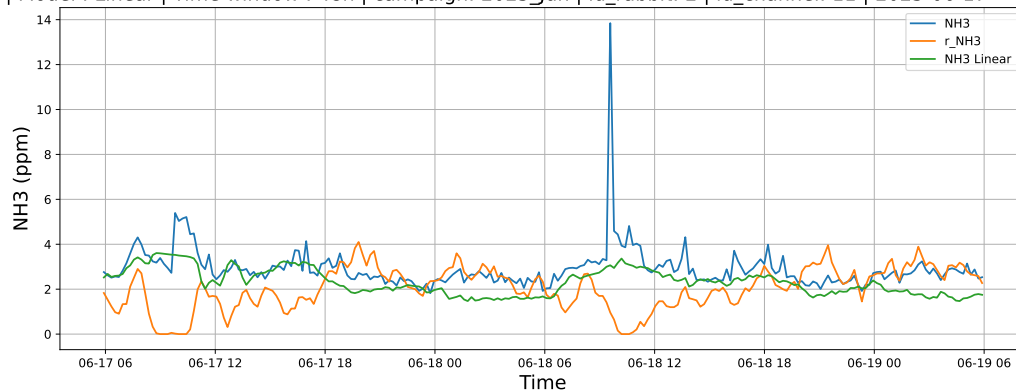


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

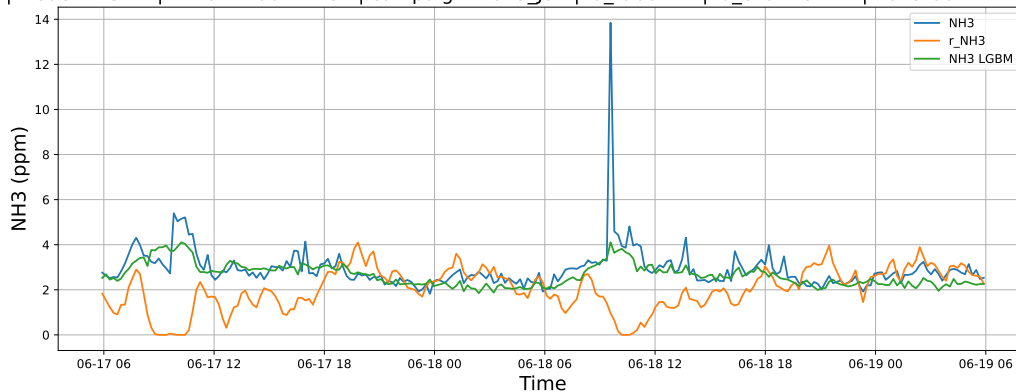


4.1.2.2 Time window : 48

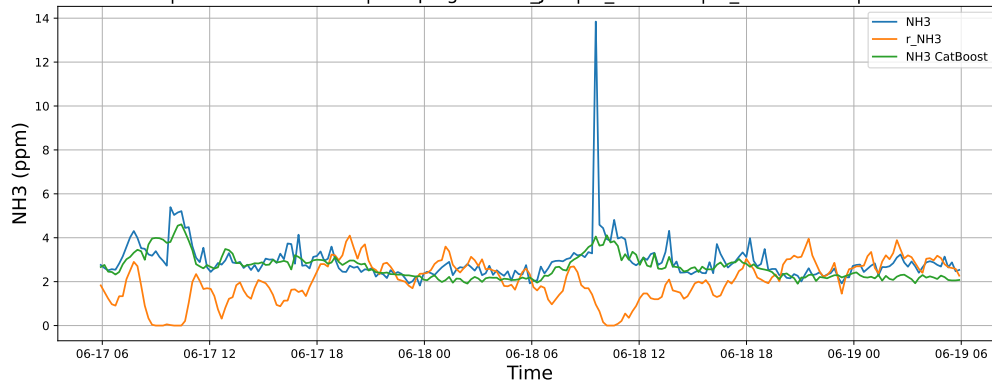
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

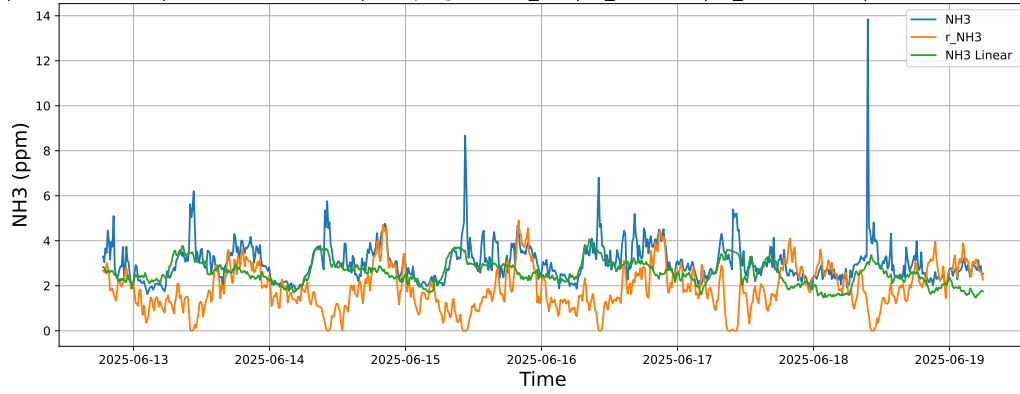


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

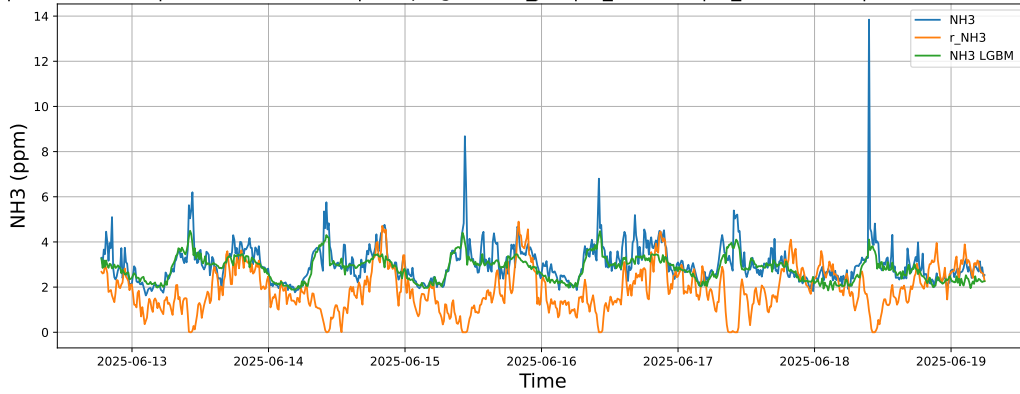


4.1.2.3 Time window : ALL

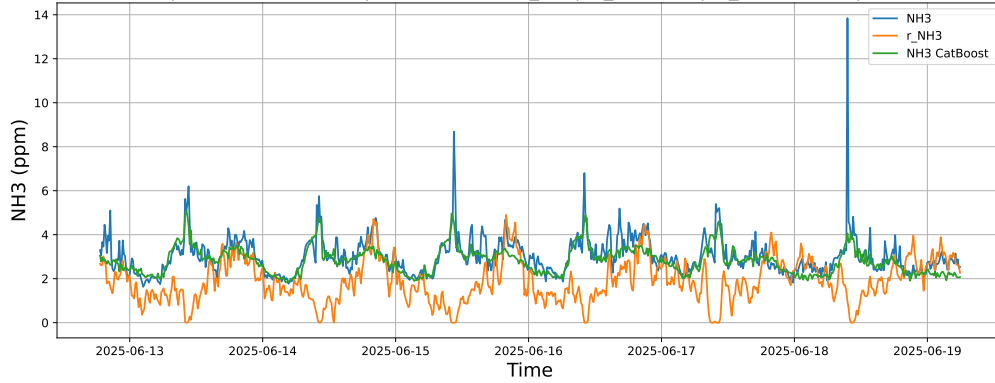
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



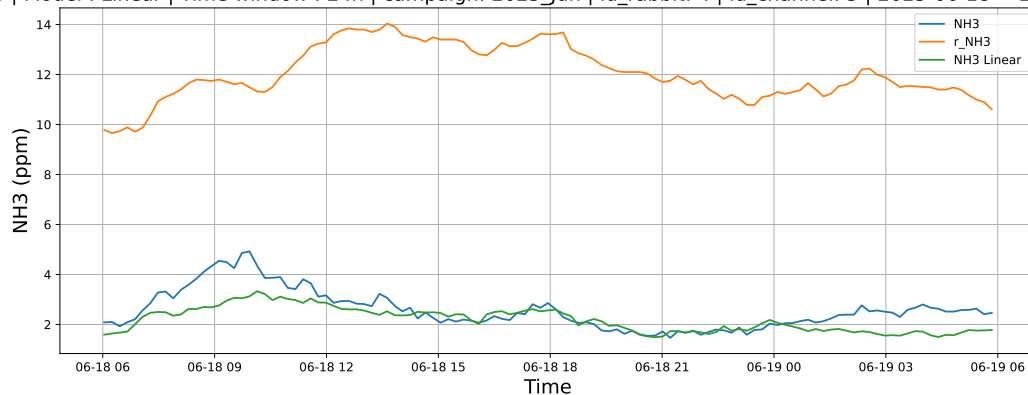
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



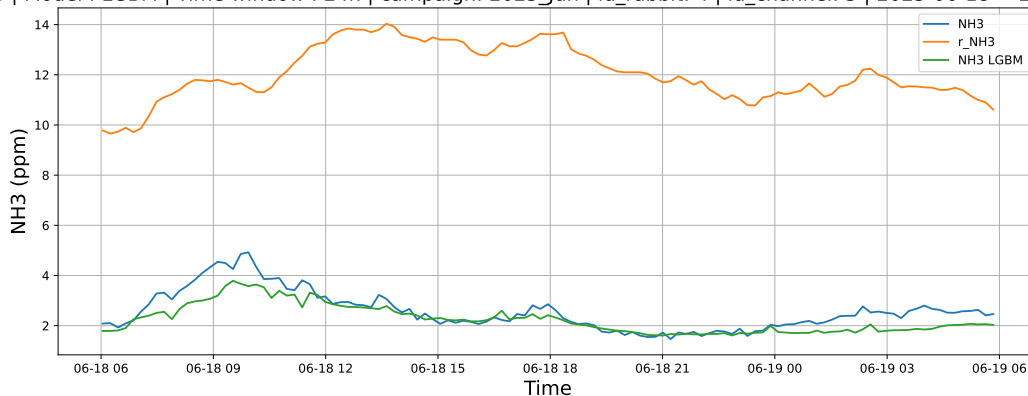
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

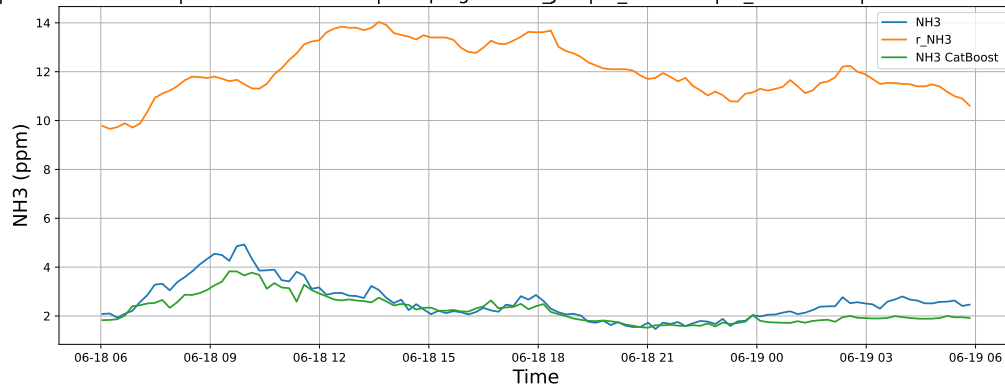
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

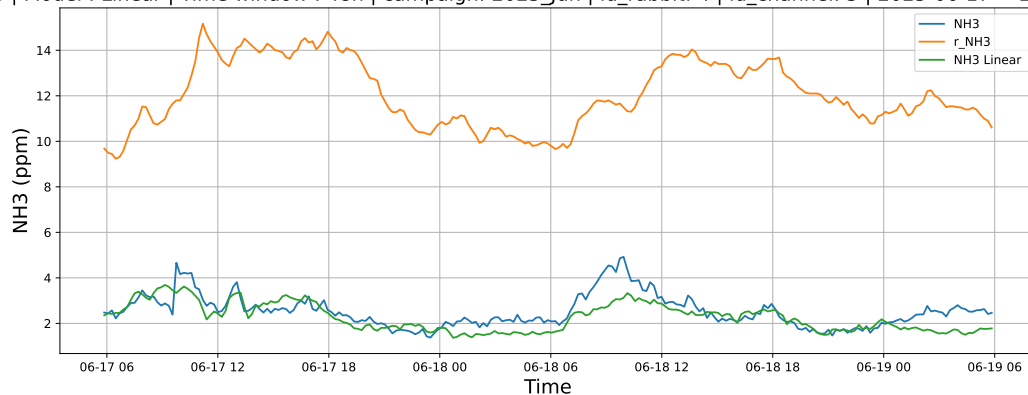


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

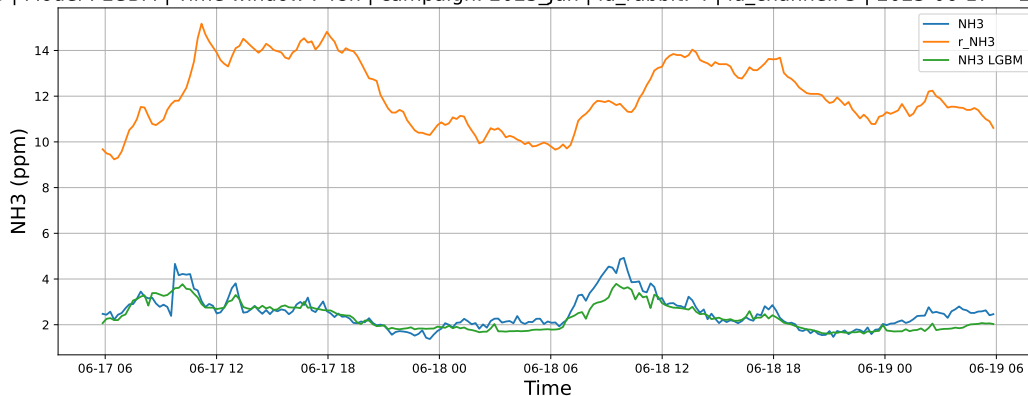


4.1.3.2 Time window : 48

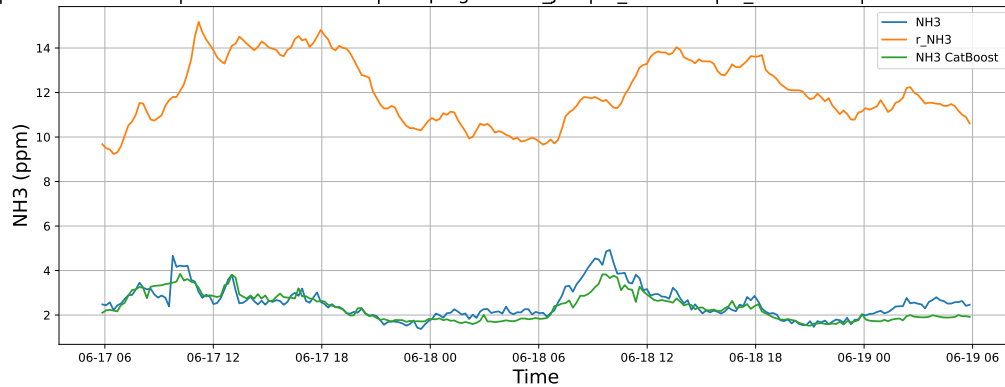
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

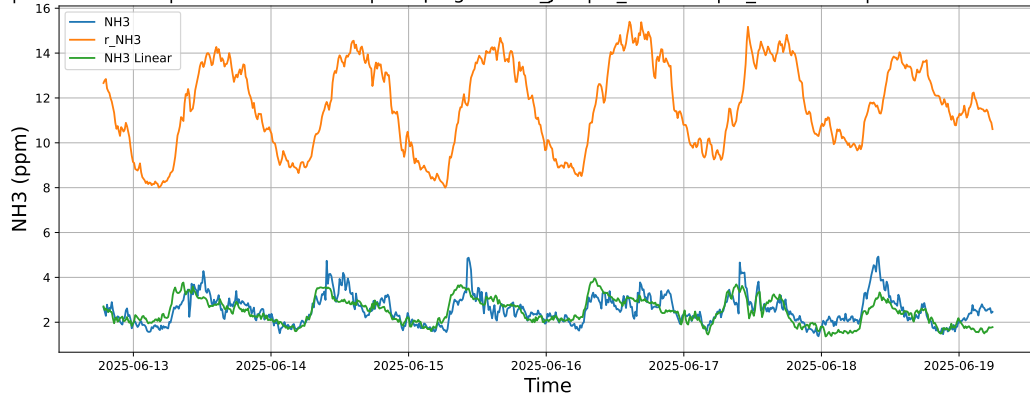


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

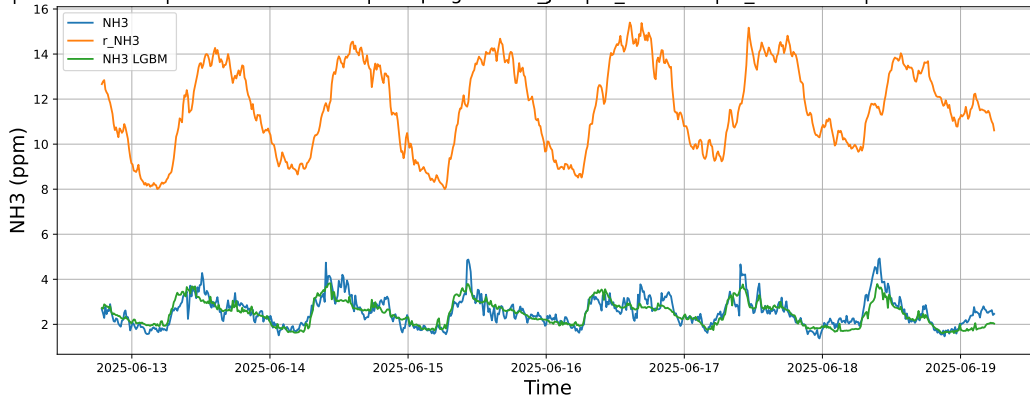


4.1.3.3 Time window : ALL

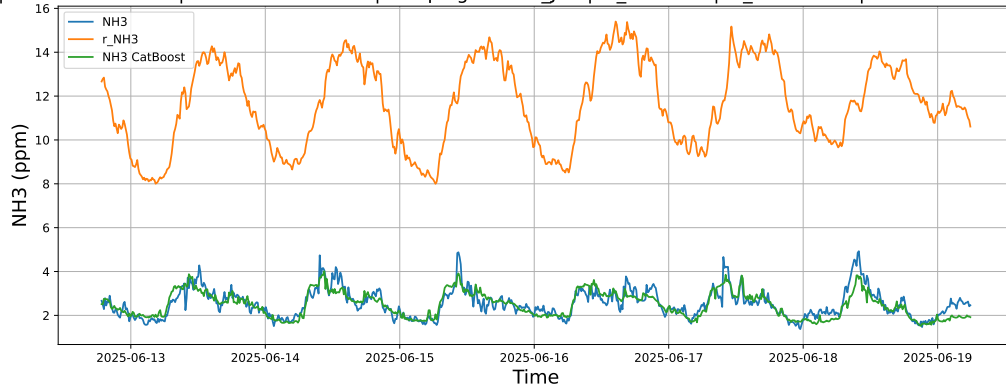
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



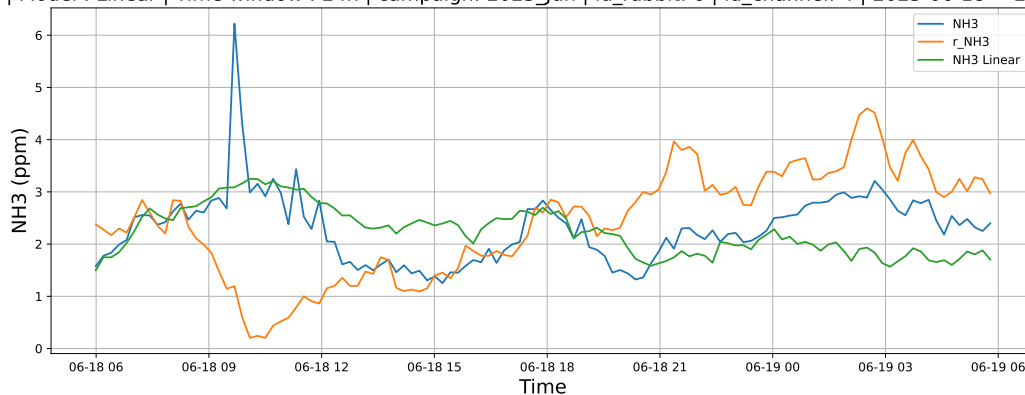
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

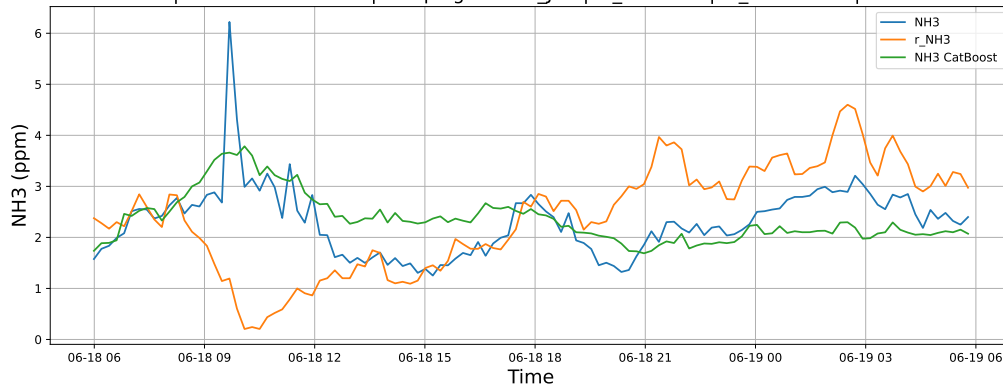
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

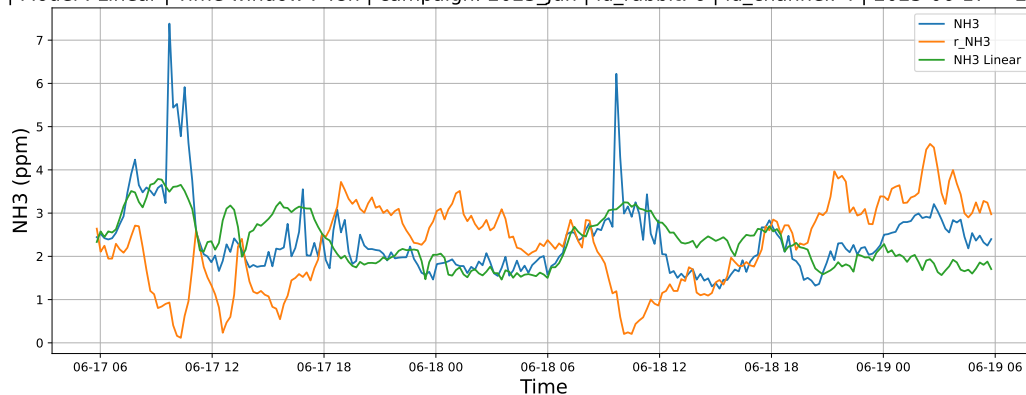


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

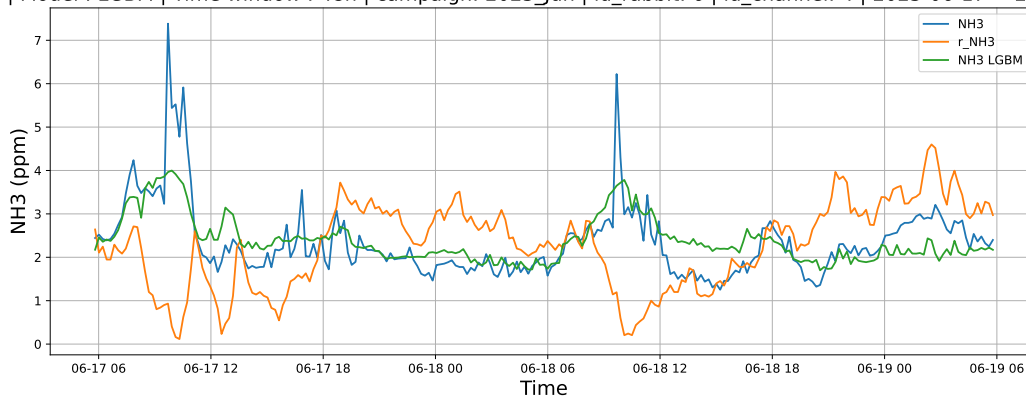


4.1.4.2 Time window : 48

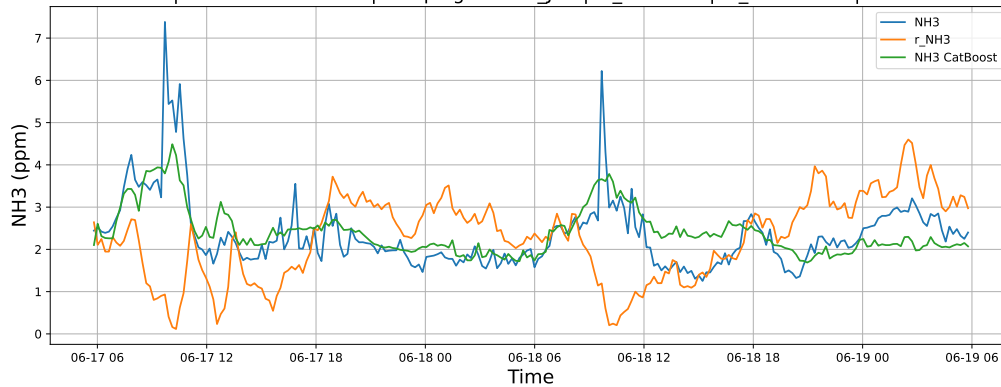
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

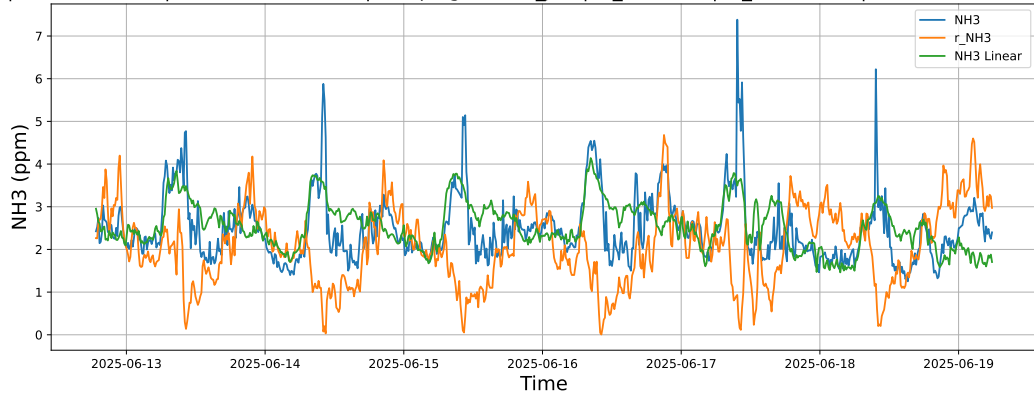


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

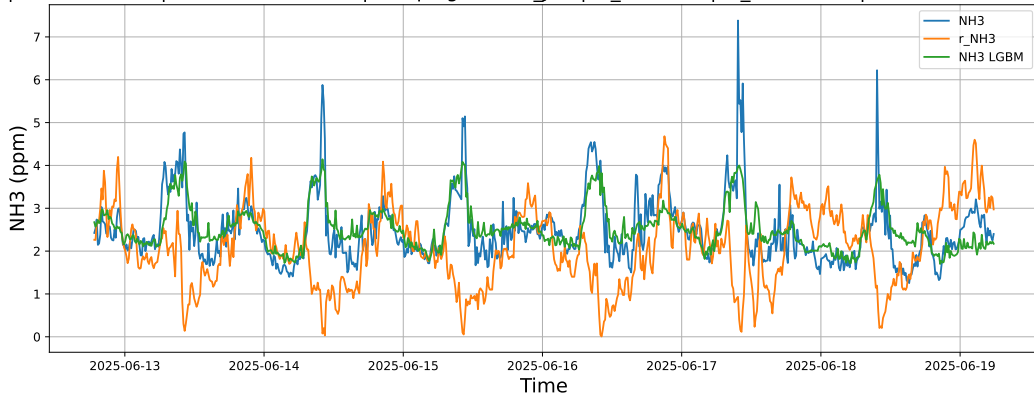


4.1.4.3 Time window : ALL

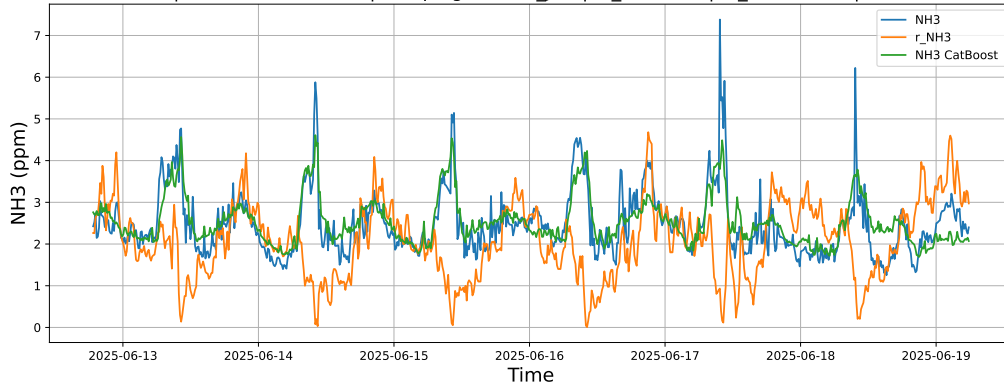
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



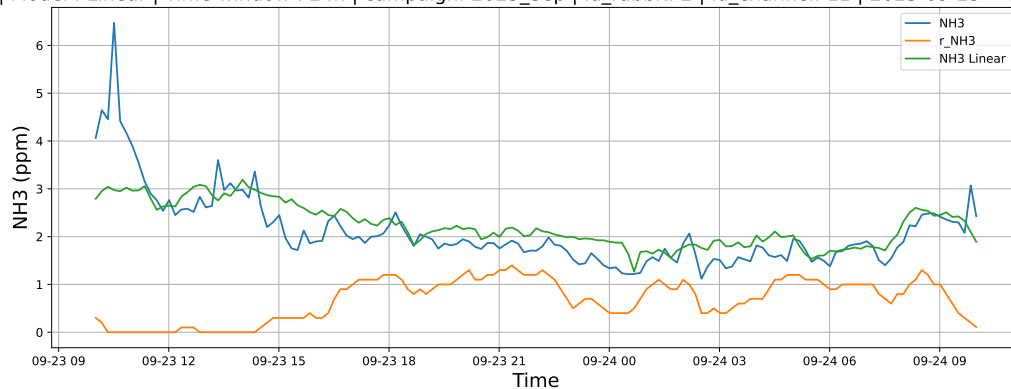
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



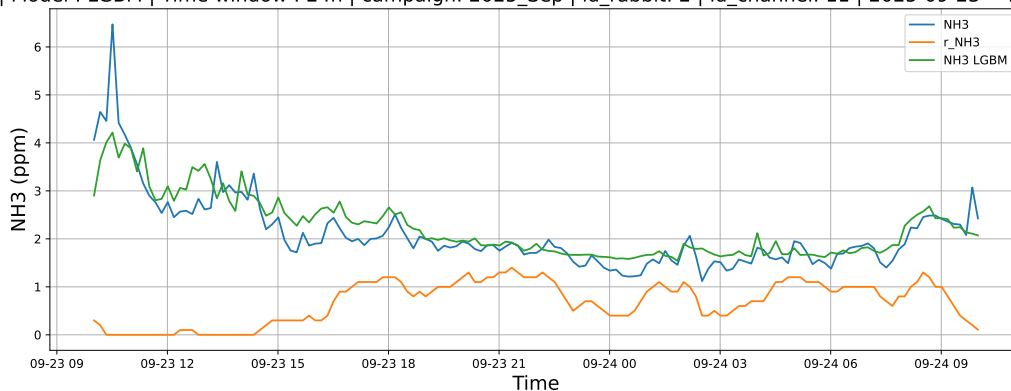
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

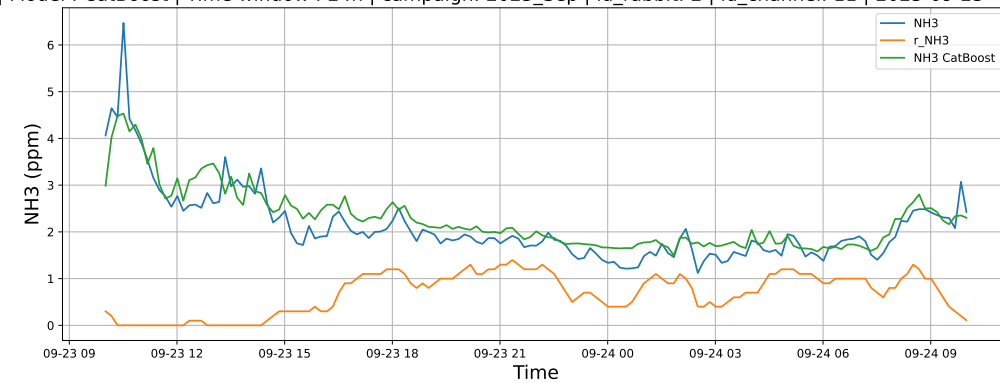
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

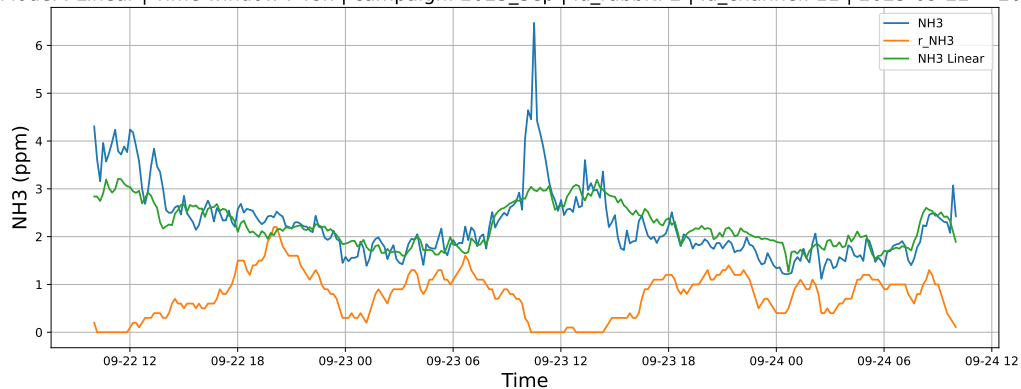


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

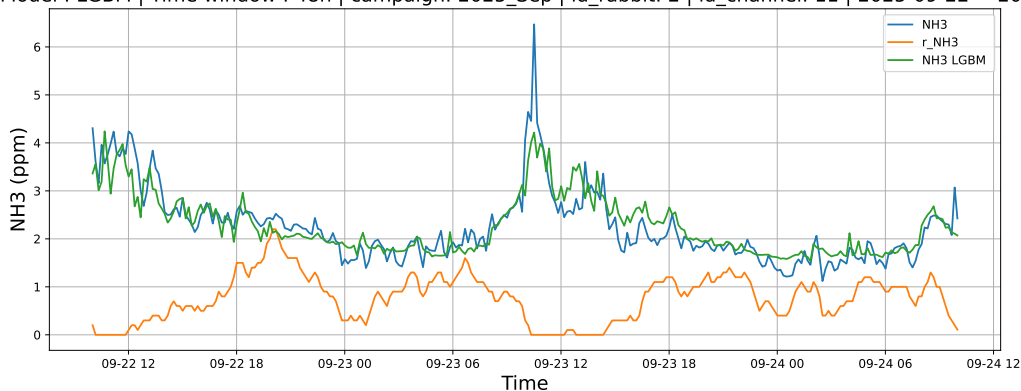


4.1.5.2 Time window : 48

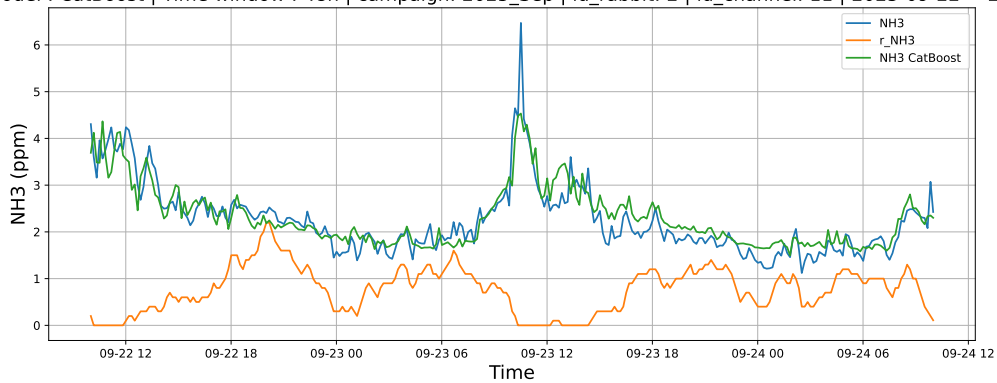
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

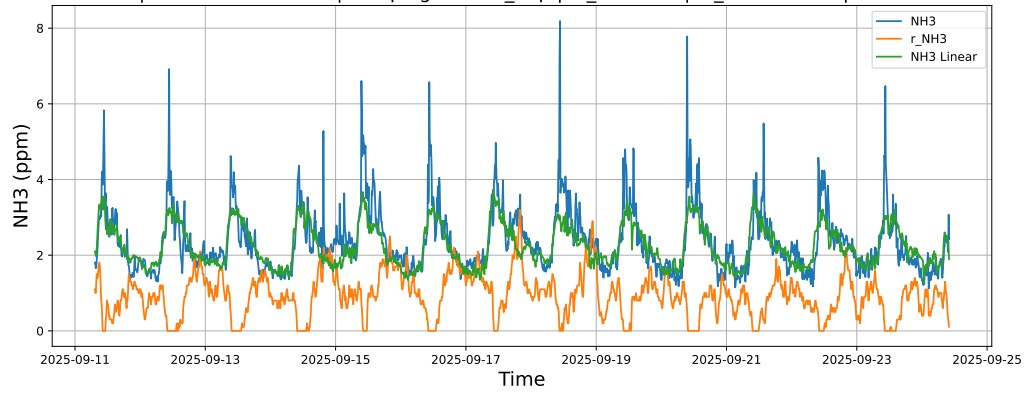


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

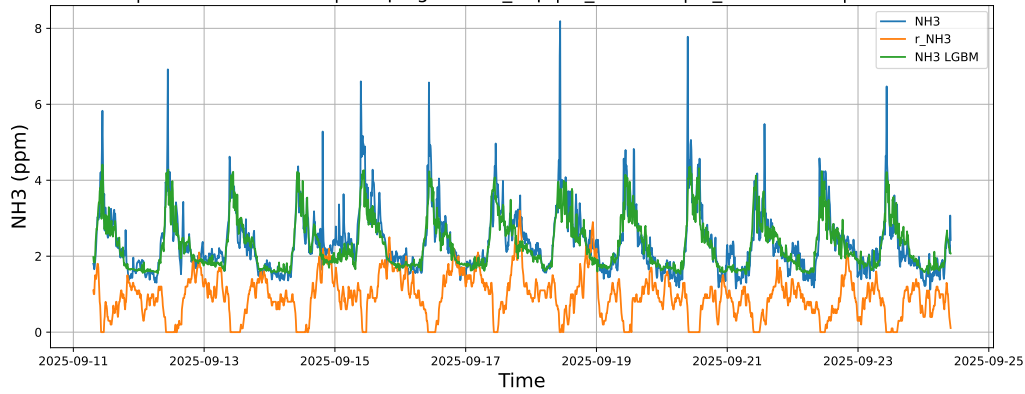


4.1.5.3 Time window : ALL

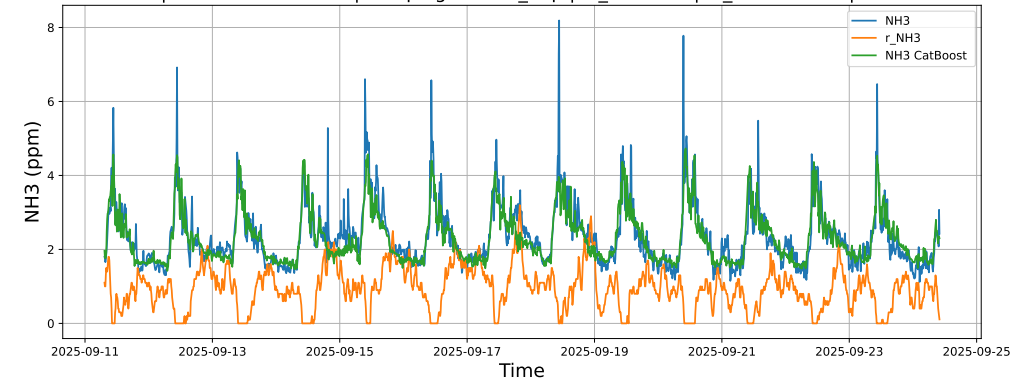
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



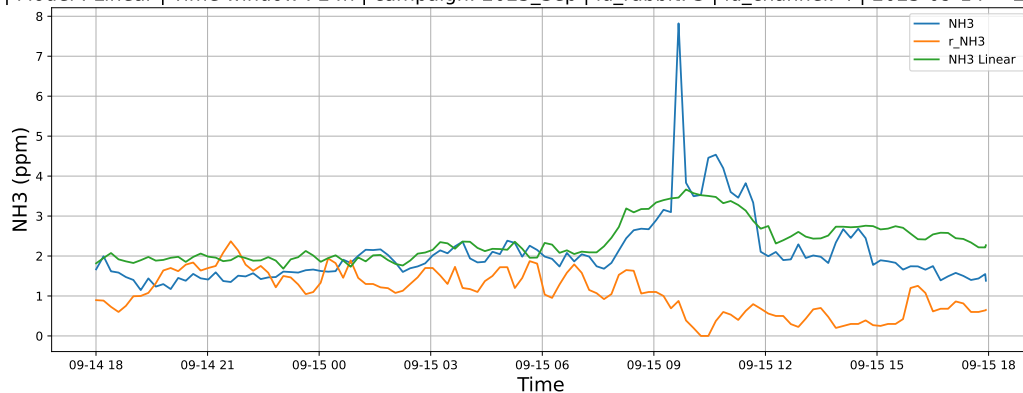
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



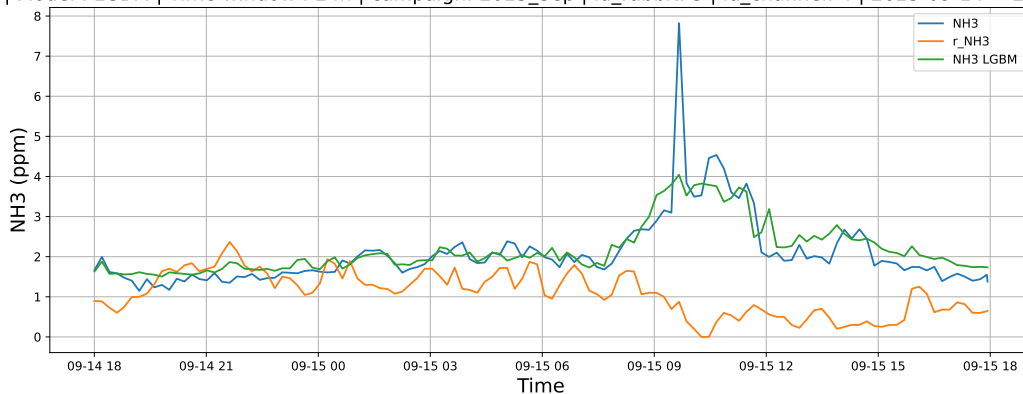
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

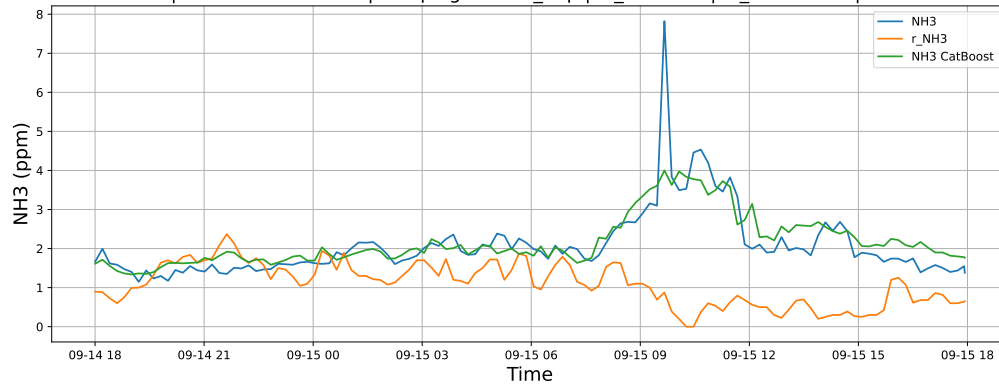
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

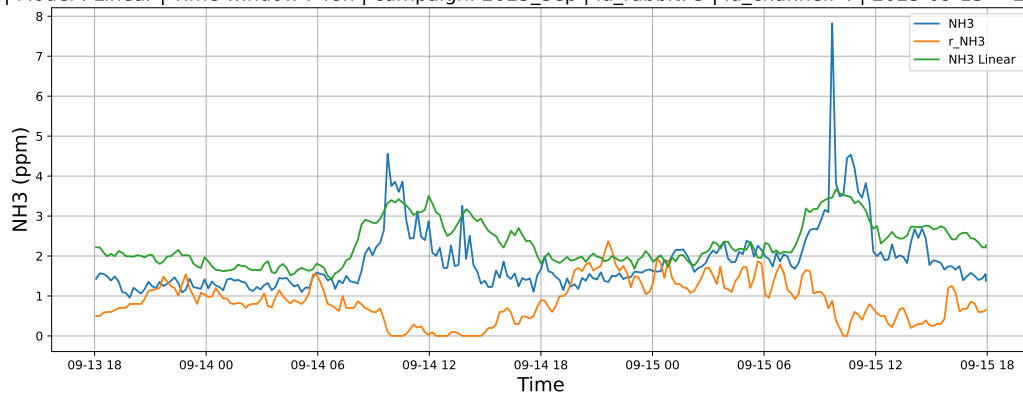


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

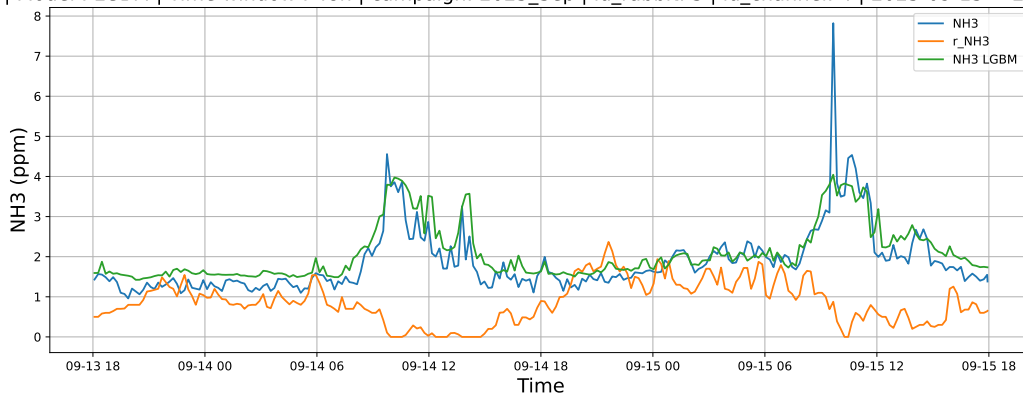


4.1.6.2 Time window : 48

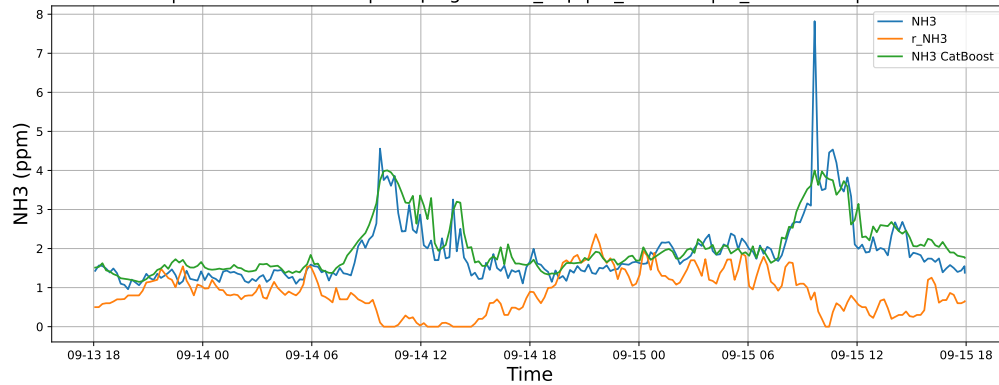
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

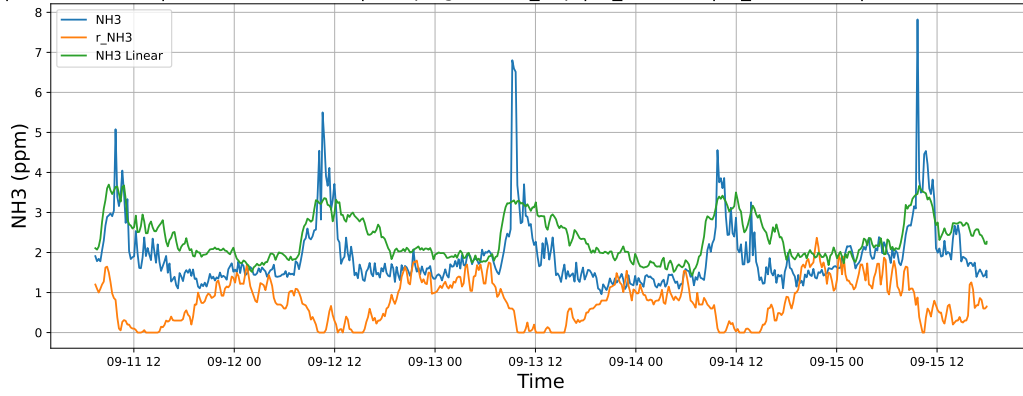


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

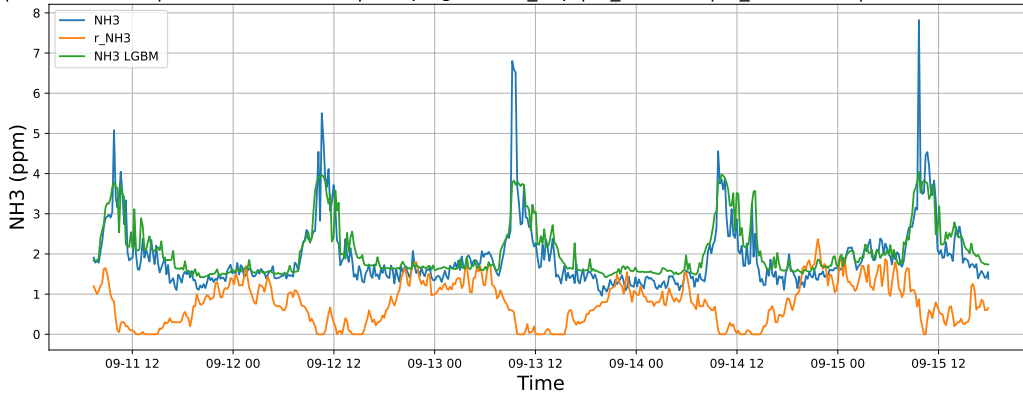


4.1.6.3 Time window : ALL

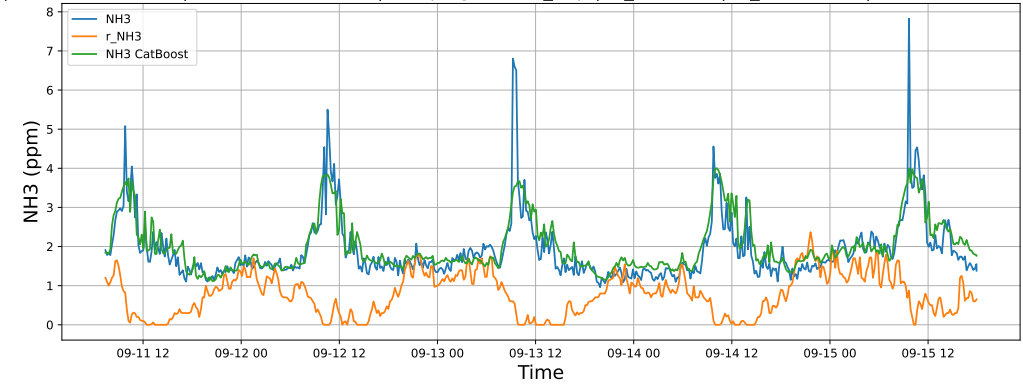
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



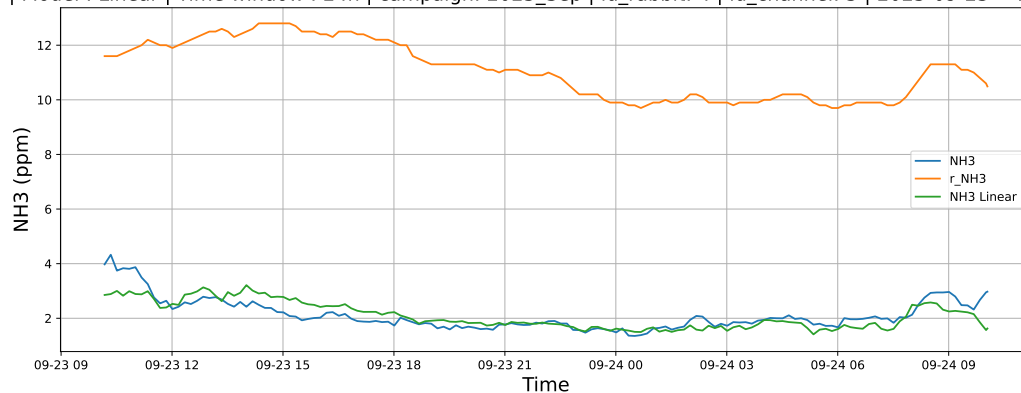
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



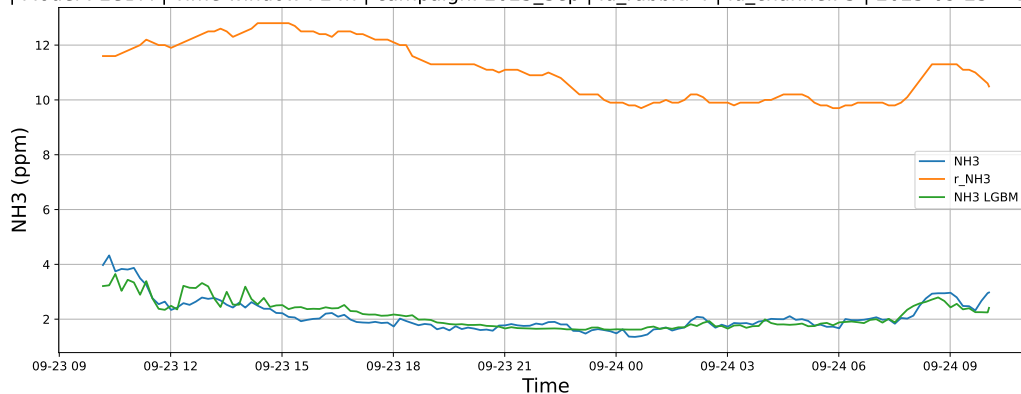
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

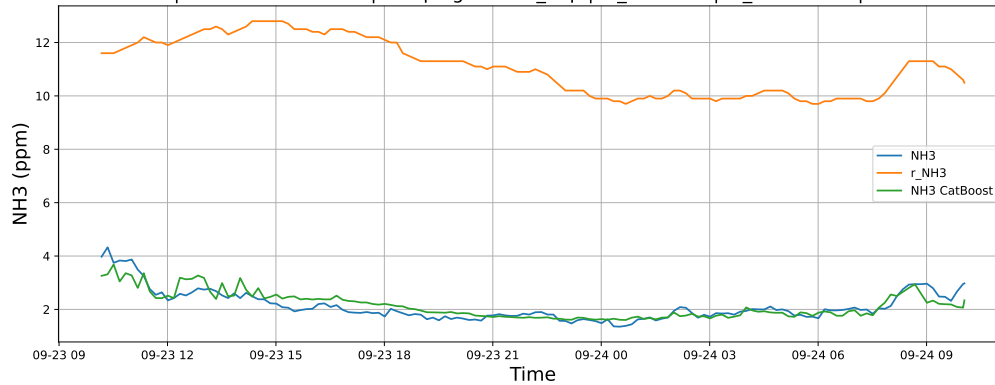
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

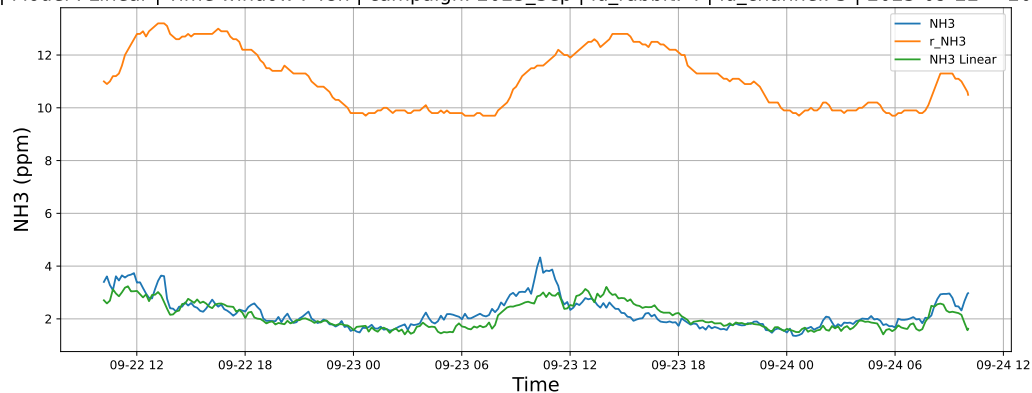


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

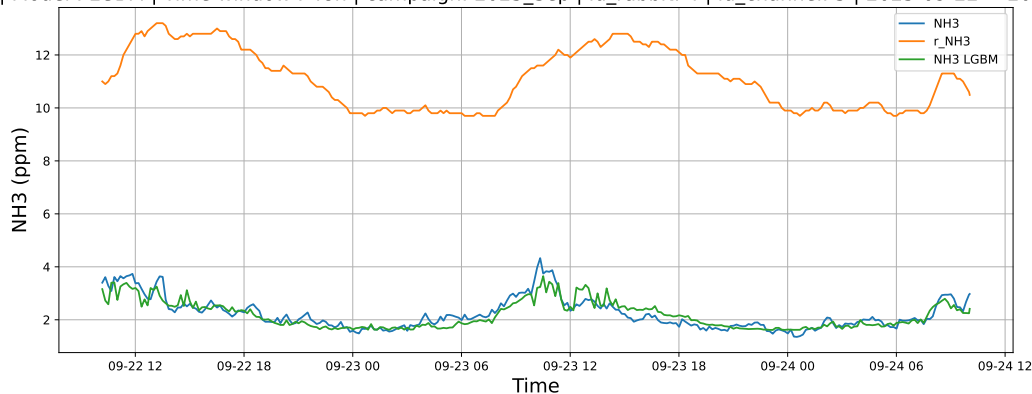


4.1.7.2 Time window : 48

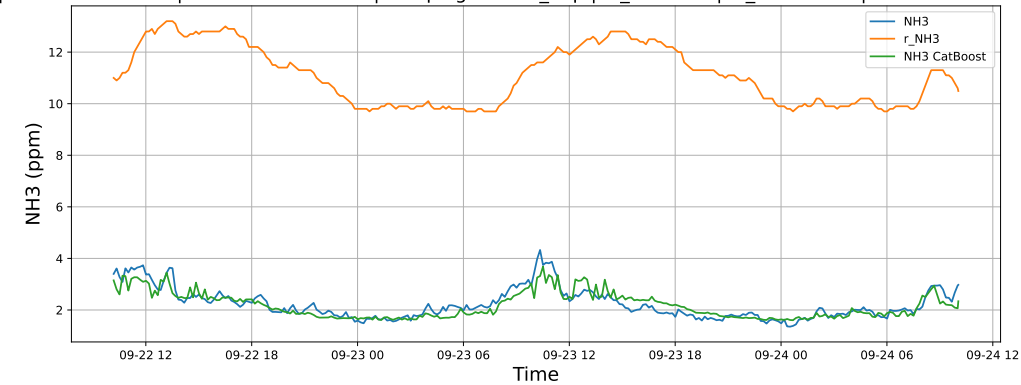
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

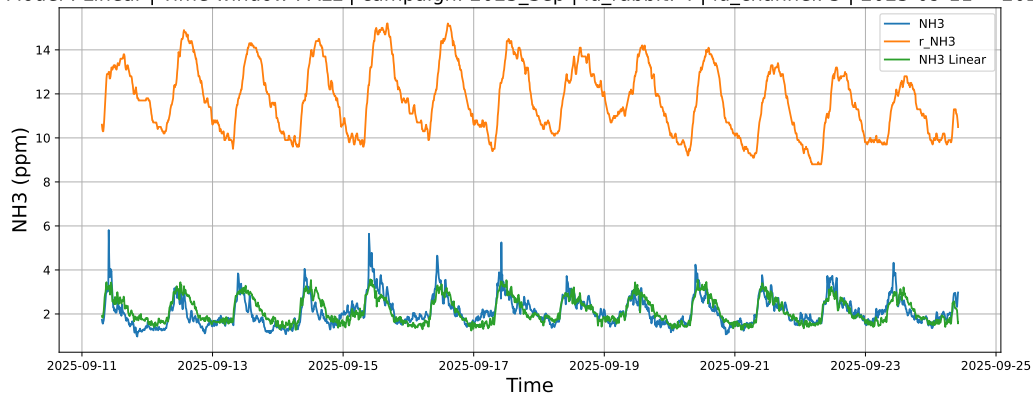


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

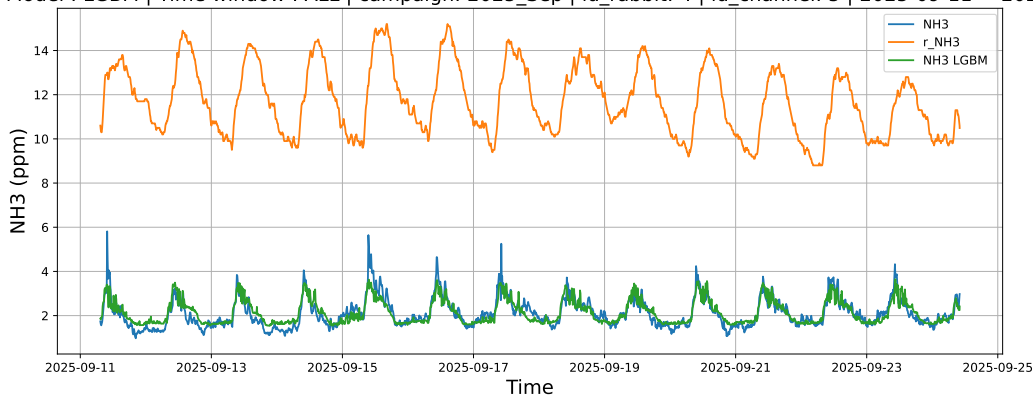


4.1.7.3 Time window : ALL

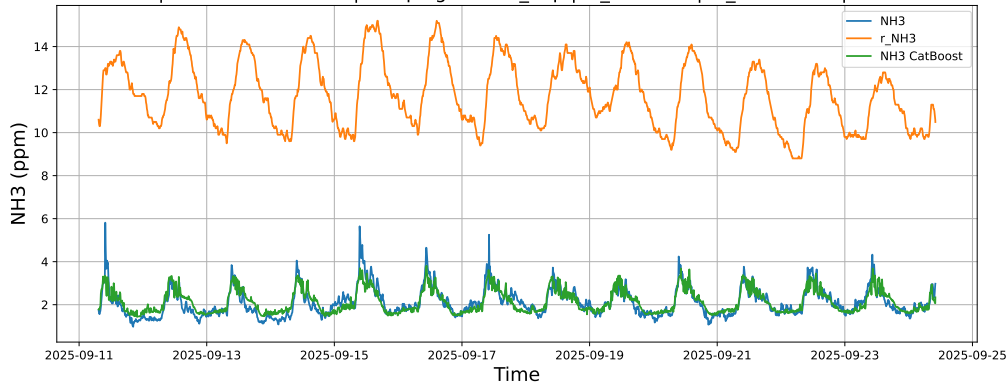
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



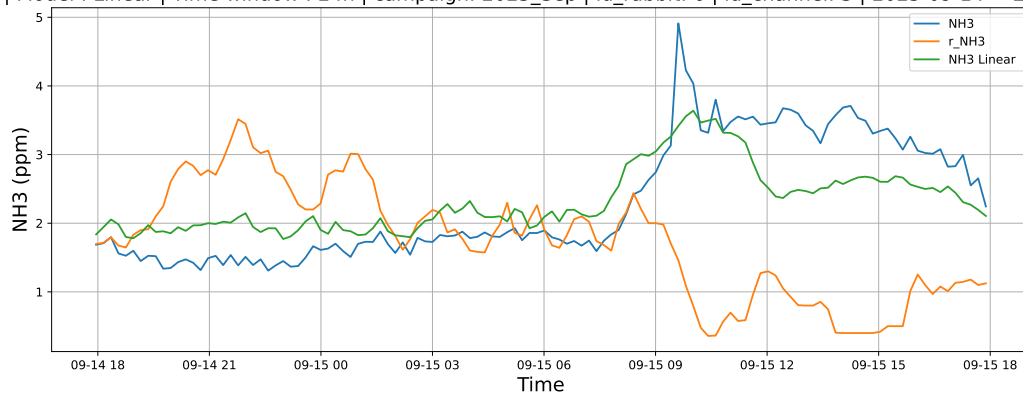
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



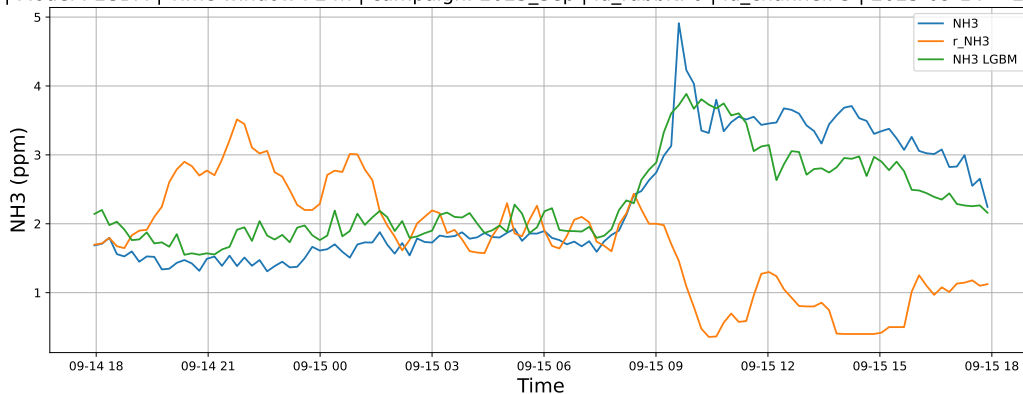
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

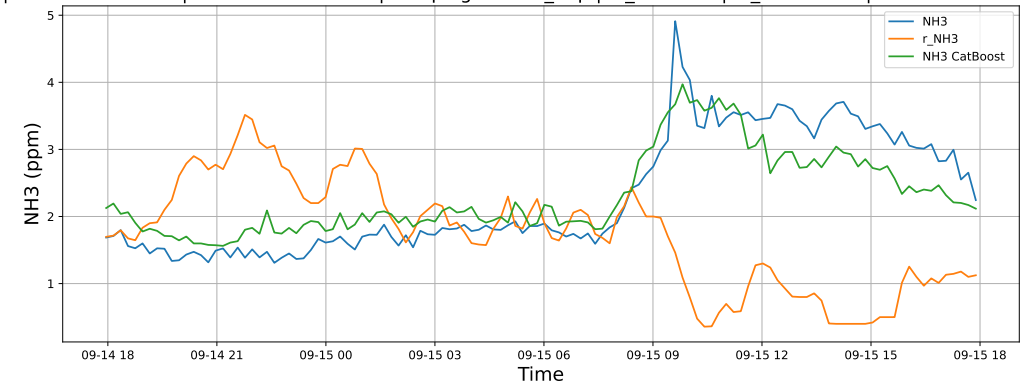
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

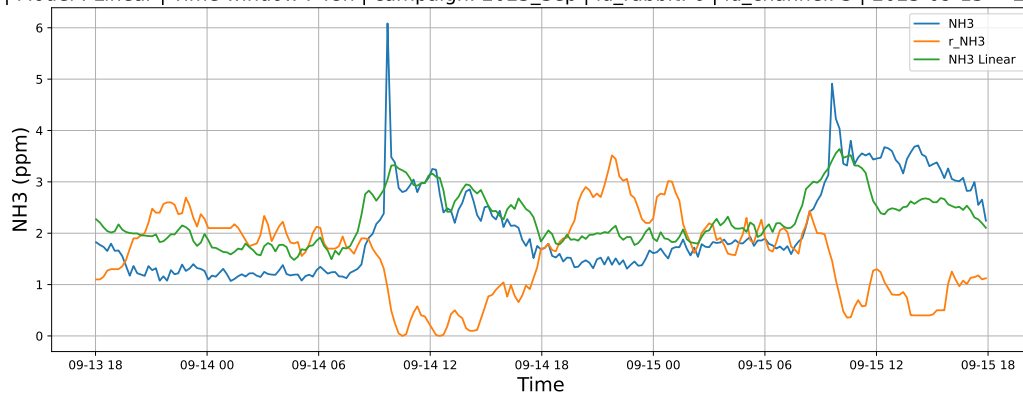


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

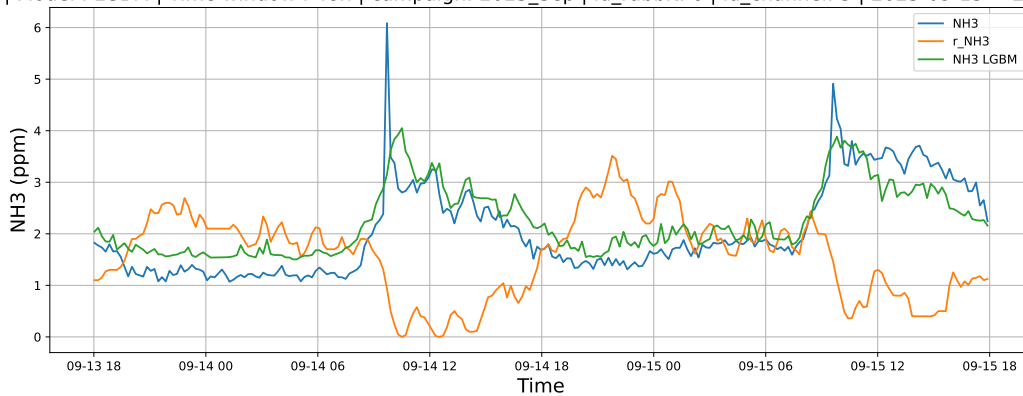


4.1.8.2 Time window : 48

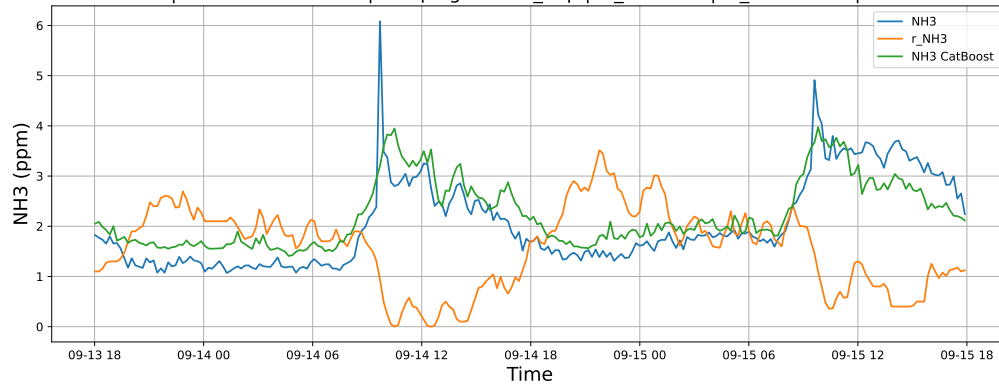
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

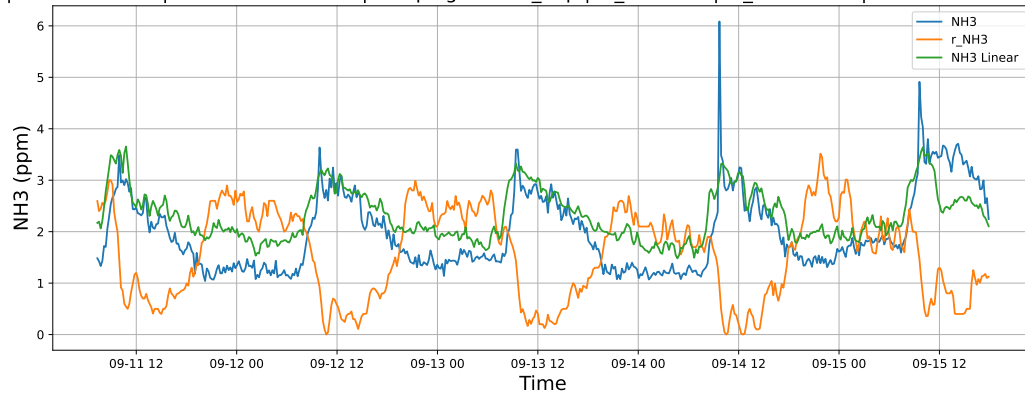


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

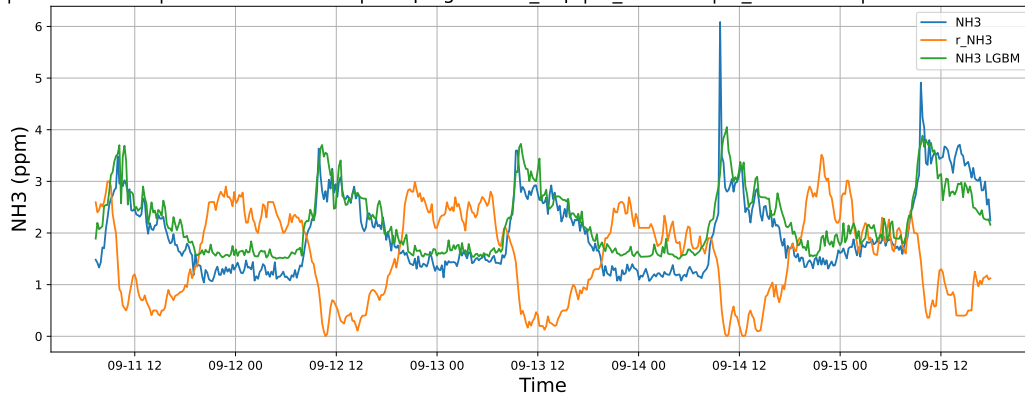


4.1.8.3 Time window : ALL

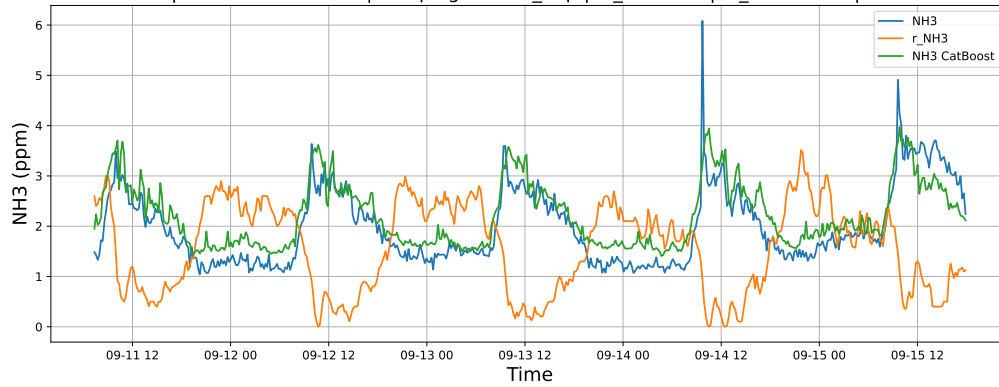
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

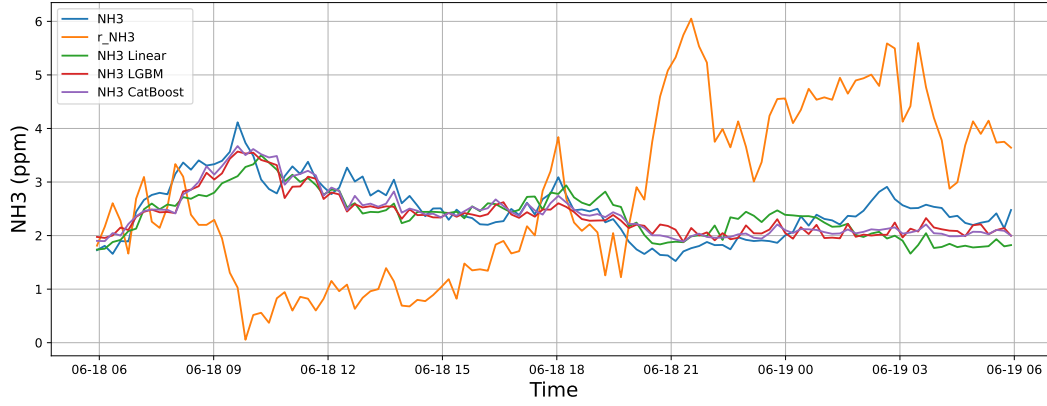


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

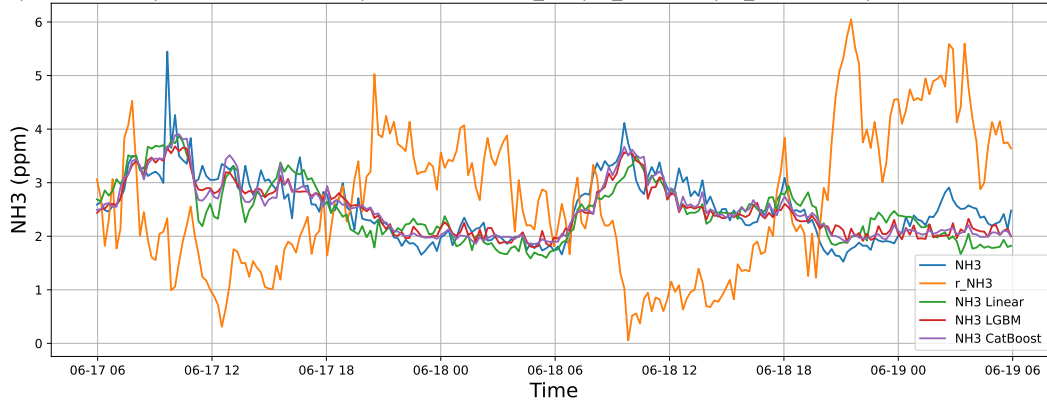
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



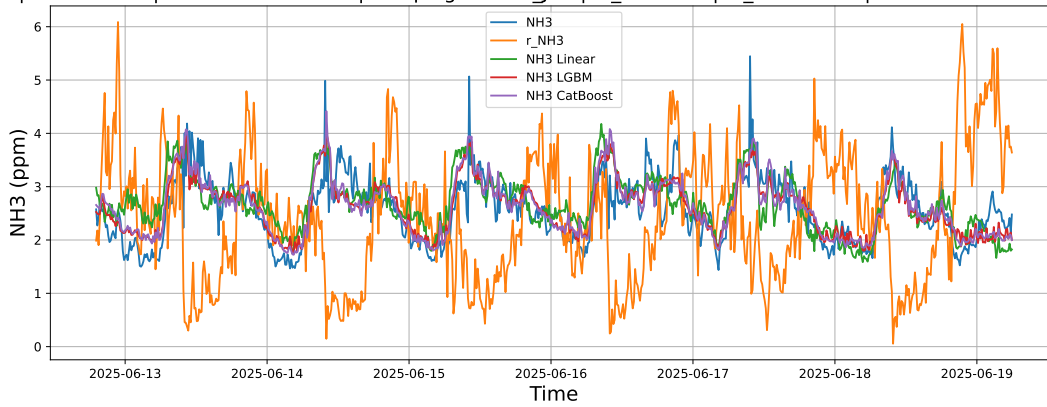
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

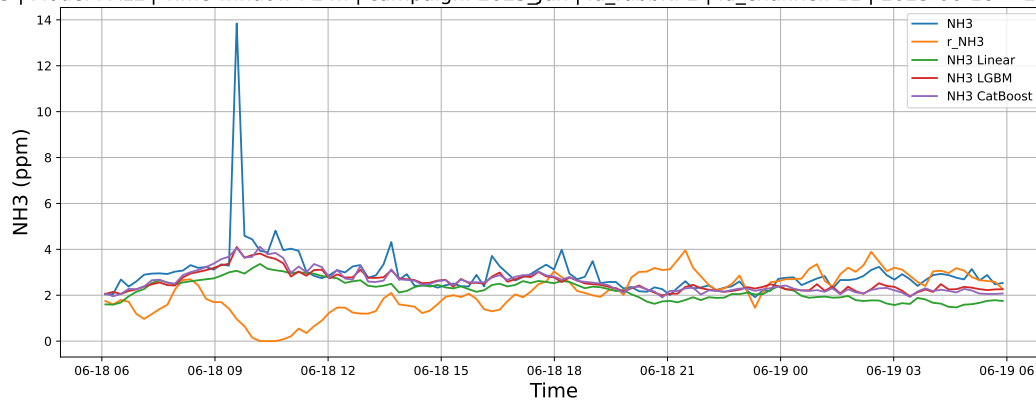
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

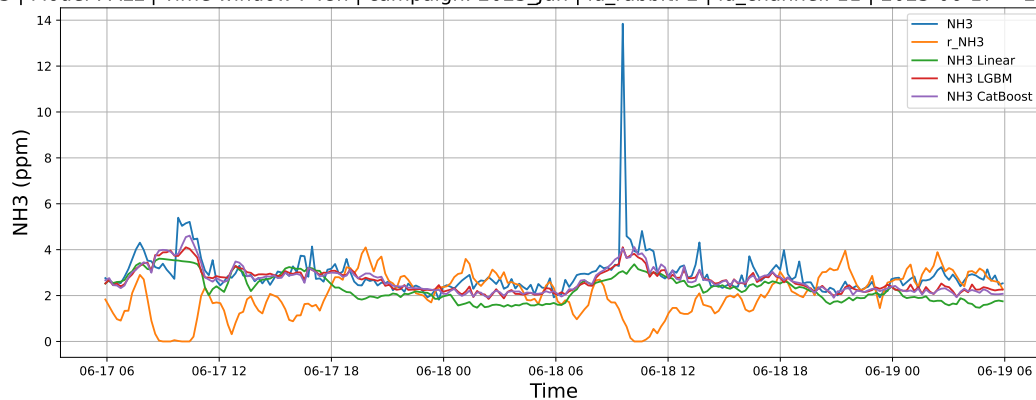
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



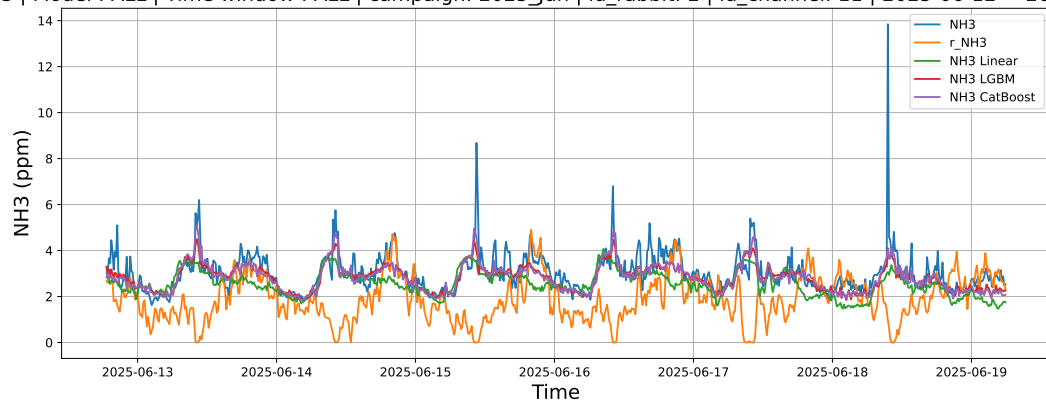
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

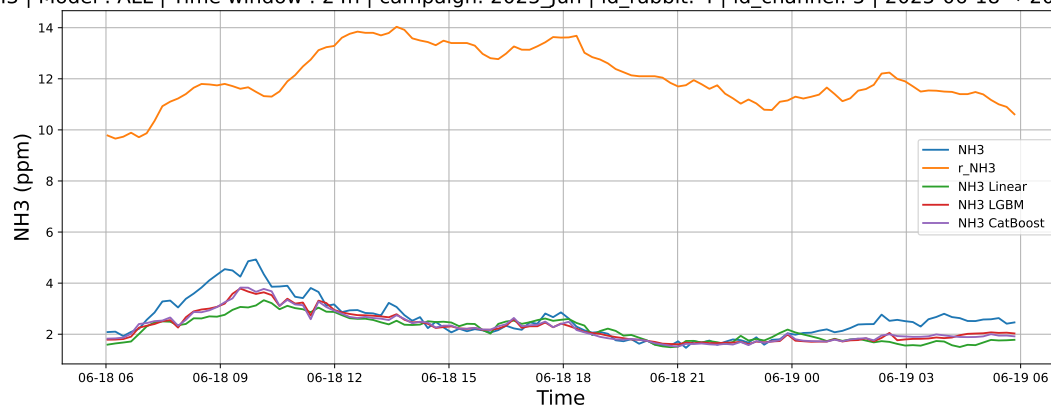
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

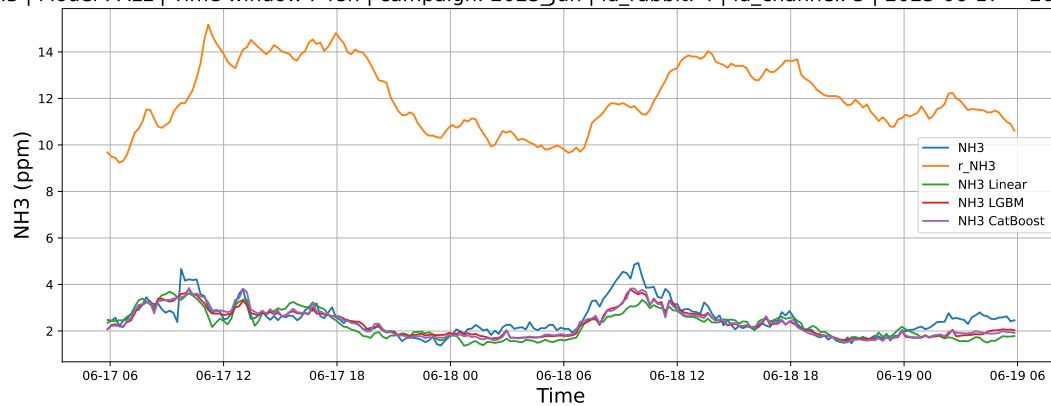
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



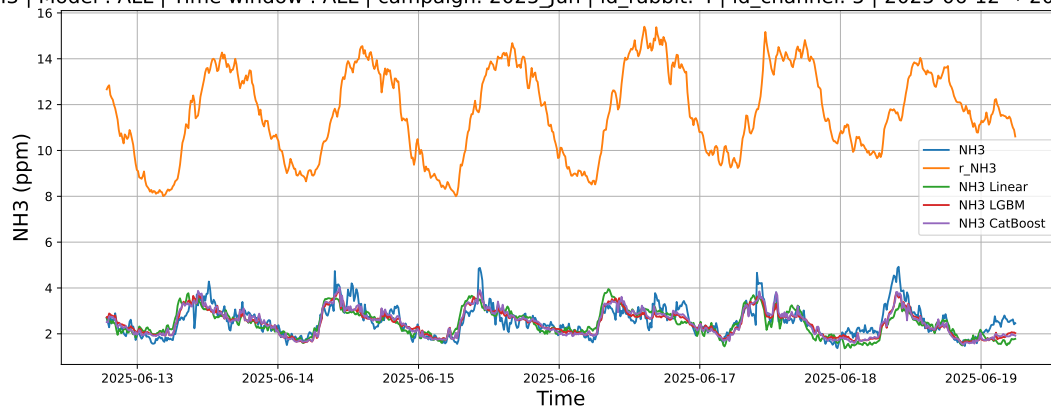
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

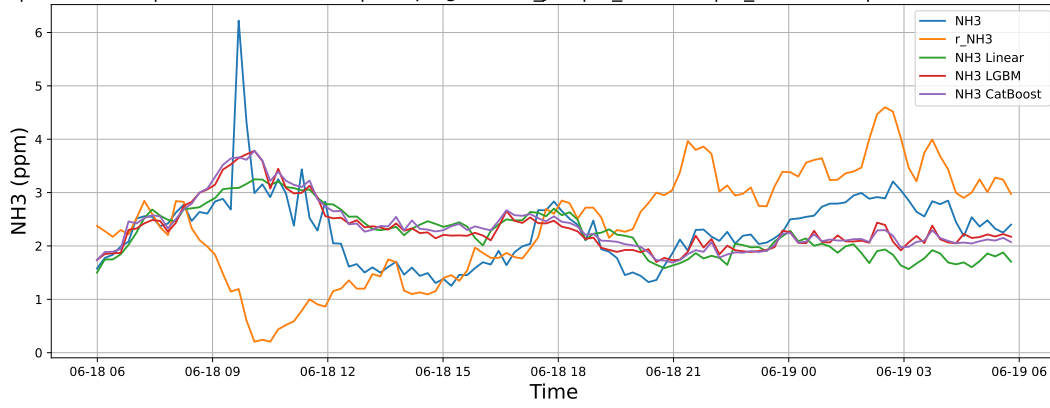
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

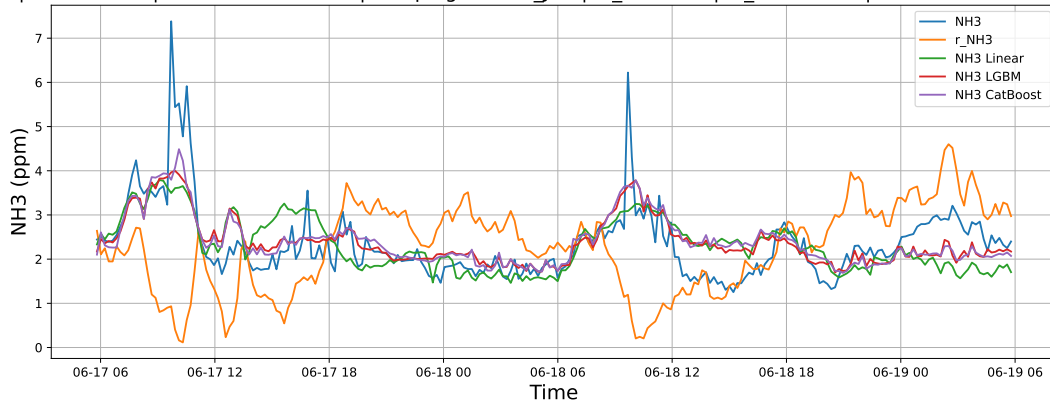
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



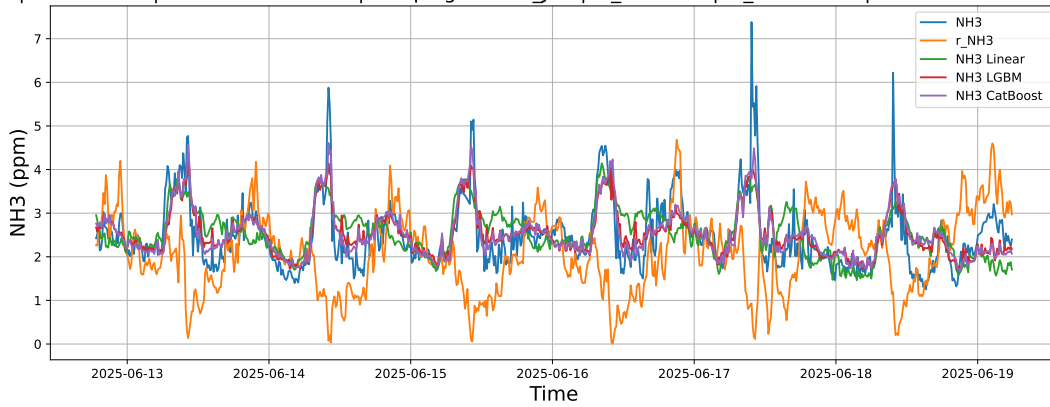
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

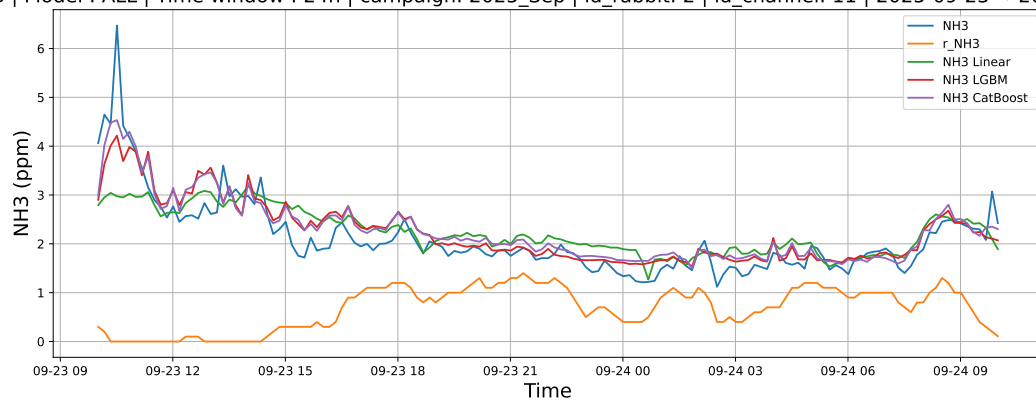
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

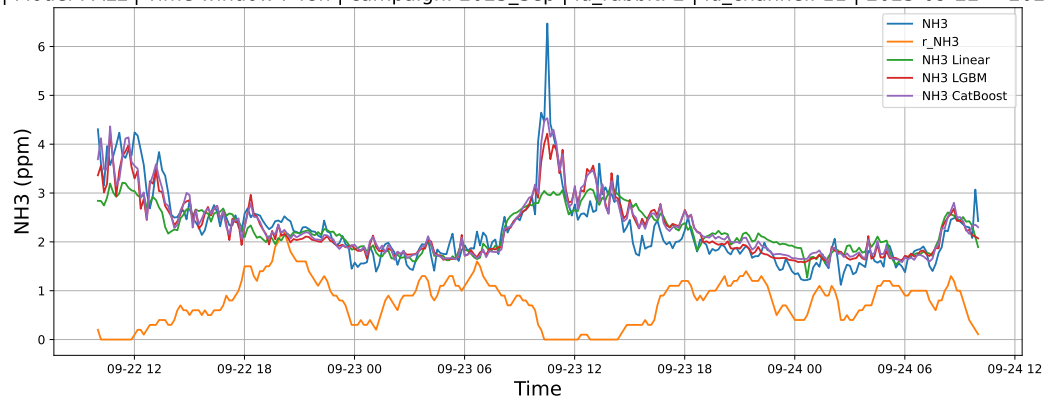
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



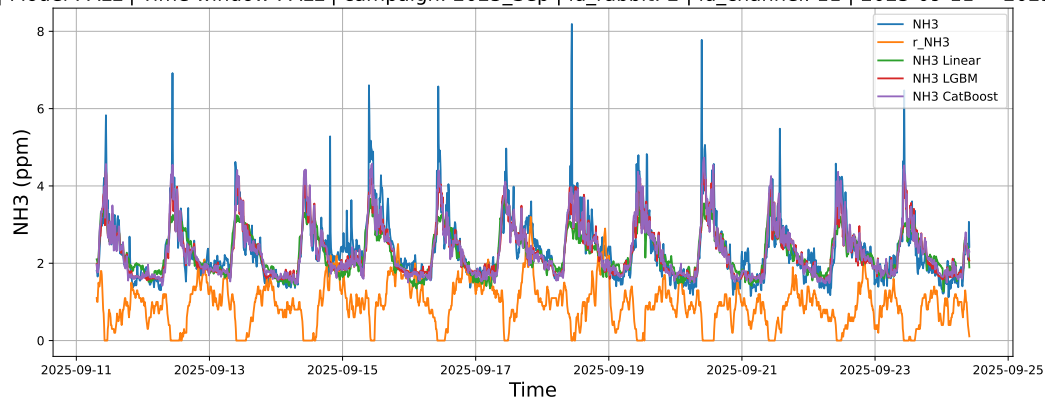
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

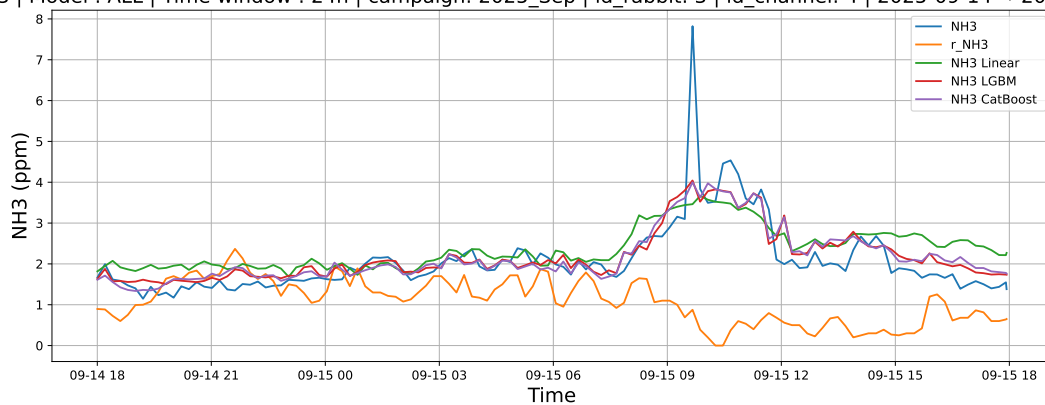
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

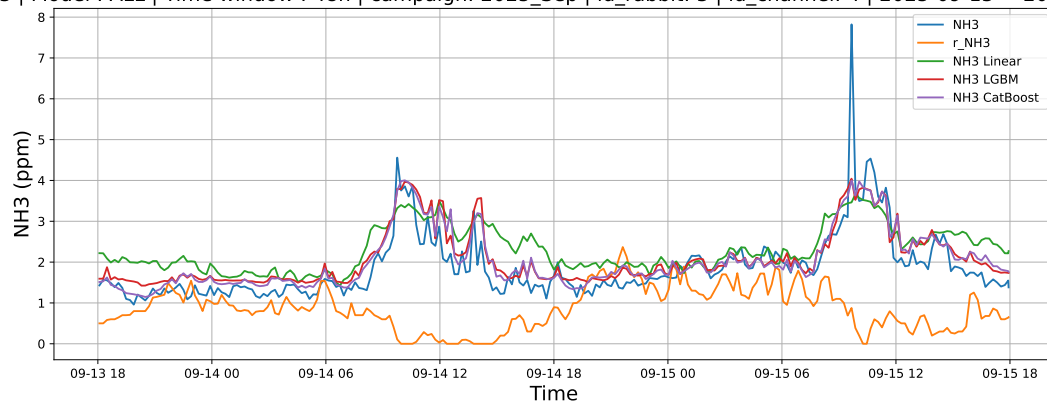
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



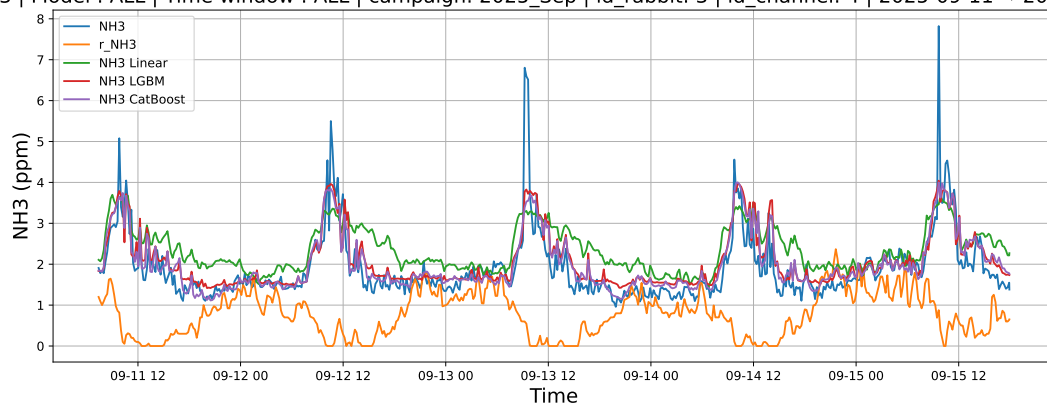
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

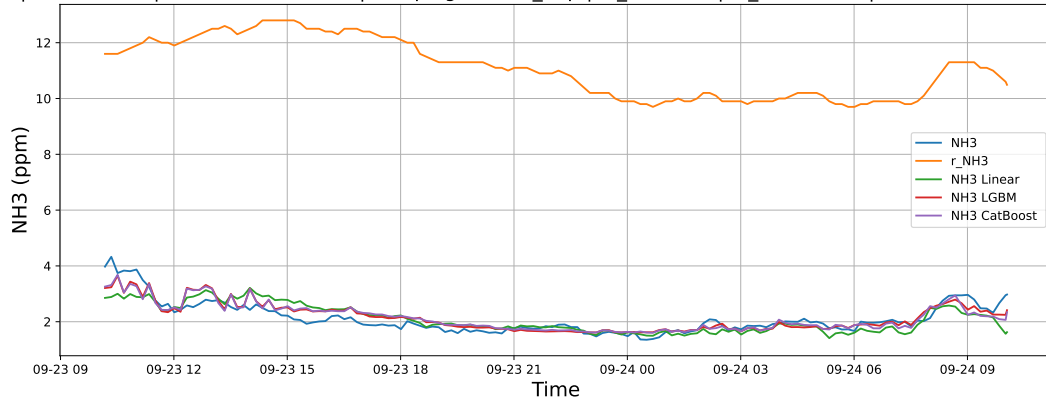
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

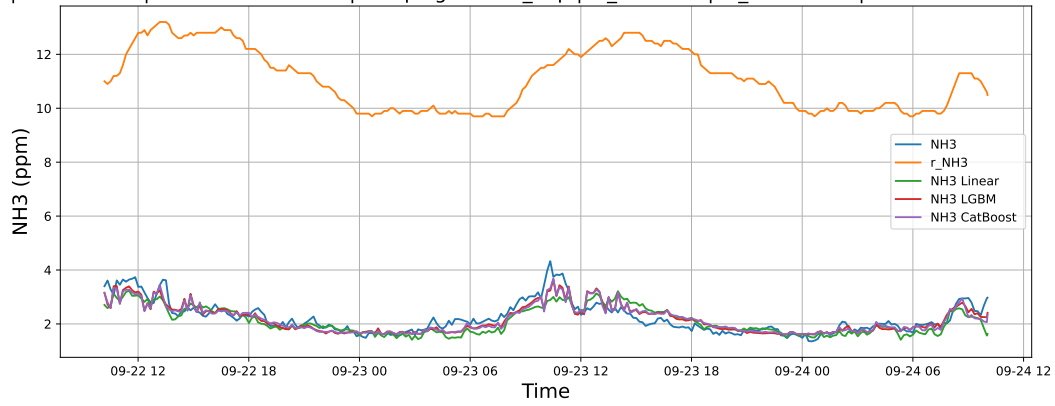
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



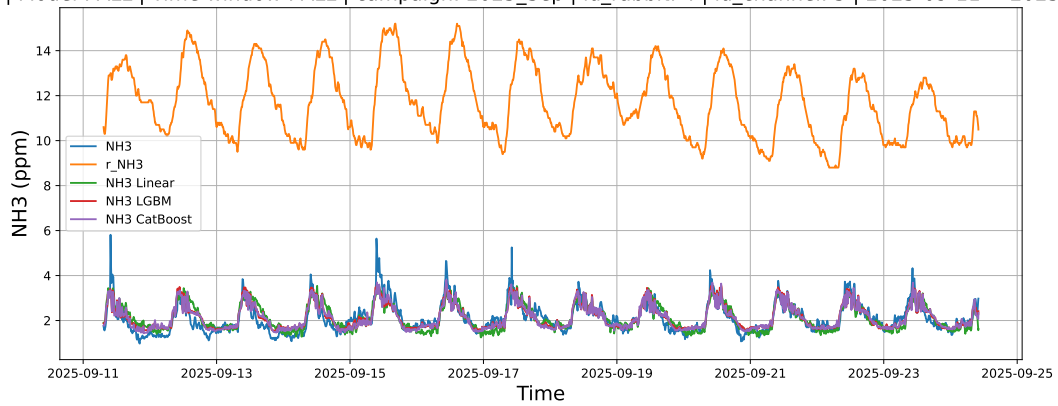
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

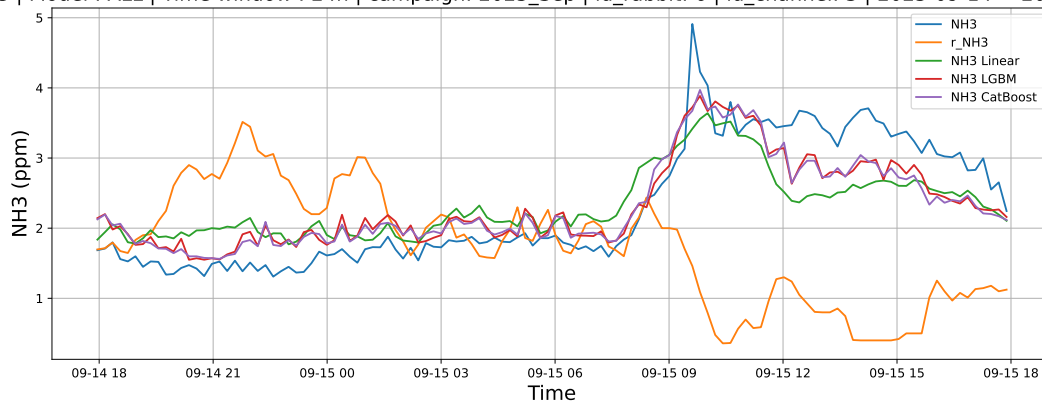
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

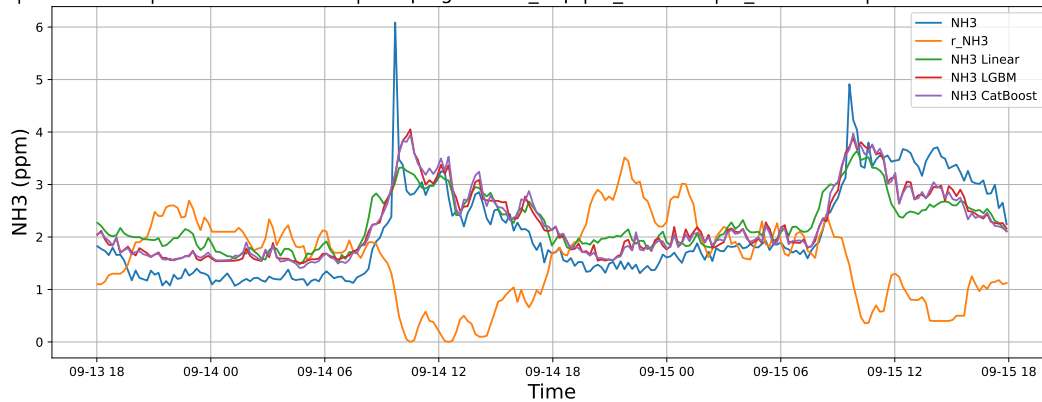
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

